MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

A body has maximum range R_1 when projected up the inclined plane. It has the maximum range R_2 when projected down the plane with same speed as in first case. Then which of the following is maximum range (R) for it when projected at certain angle with the horizontal ground and with same speed as in previous two cases?

(A) $R = R_1 + R_2$

(B) $R = R_2 - R_1$

(C) $R = \frac{R_1 R_2}{R_1 + R_2}$

(D) $R = \frac{2R_1 R_2}{R_1 + R_2}$

Q.2

A lift is moving upwards with a uniform velocity v in which a block of mass m is lying. The frictional force offered by the block, when coefficient of the frictional force is μ, will be

(A) zero

(B) mg

(C) μ mg

(D) 2μ mg

Q.3

Assertion: Our ears cannot distinguish two notes, one produced by a violin and other by a sitar, if they have exactly the same intensity and same frequency.
Reason: When a musical instrument is played, it produces a fundamental note which is accompanied by number of overtones called harmonics.

(A) Assertion is true, reason is true; reason is a correct explanation for assertion.

(B) Assertion is true, reason is true; reason is not a correct explanation for assertion.

(C) Assertion is true, reason is false.

(D) Assertion is false but, reason is true.

Q.4

Assertion: In the Maxwellian distribution of speeds in molecules of gases, no. of molecules having speed less than v_{av} is lesser than number of molecules having speed greater than v_{av}.
Reason: Area under $\frac{dN}{du}$ and u curve is more for molecules having speed greater than v_{av}.

(A) If both assertion and reason are true and the reason is correct explanation of the assertion.

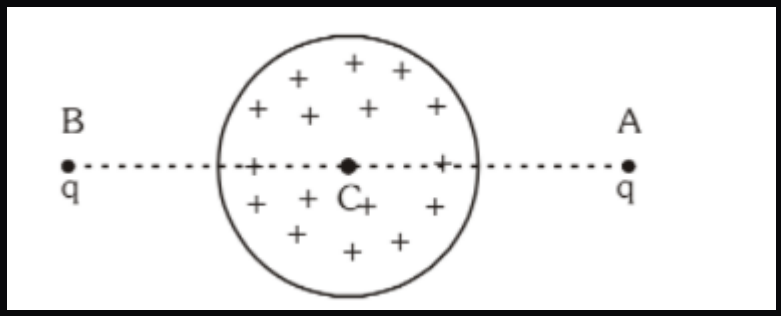
(B) If both assertion and reason are true but reason is not correct explanation of the assertion.

(C) If assertion is true, but the reason is false.

(D) If assertion is false, but reason is true.

Q.5

A uniformly charged non-conducting sphere having charge Q and radius R is kept in gravity free space. Two identical charge particles are released from rest from $'A'$ and $'B'$ simultaneously. The speed of one of the charge particle when separation between them is very large, is (mass of one of the particle is m) $\left[\mathrm{C} \rightarrow\right.$ centre of sphere and A, B and C in same line. $\mathrm{AC}=\mathrm{BC}=2 \mathrm{R}, \mathrm{K}=\frac{1}{4 \pi \varepsilon_0}$]



(A) $\sqrt{\frac{\mathrm{Kq}}{2 \mathrm{mR}}\left(\frac{\mathrm{q}}{2}+2 \mathrm{Q}\right)}$

(B) $\sqrt{\frac{\mathrm{Kq}}{\mathrm{mR}}\left(\frac{\mathrm{q}}{2}+\mathrm{Q}\right)}$

(C) $\sqrt{\frac{\mathrm{Kq}^2}{2 \mathrm{mR}}}$

(D) $\sqrt{\frac{\mathrm{KqQ}}{2 \mathrm{mR}}}$

Q.6

The width of man's face is $10 \mathrm{~cm}$. The distance between the eyes of the man is $4 \mathrm{~cm}$. Then the minimum width of plane mirror to see his full face, is :

(A) $5 \mathrm{~cm}$

(B) $4 \mathrm{~cm}$

(C) $3 \mathrm{~cm}$

(D) $10 \mathrm{~cm}$

Q.7

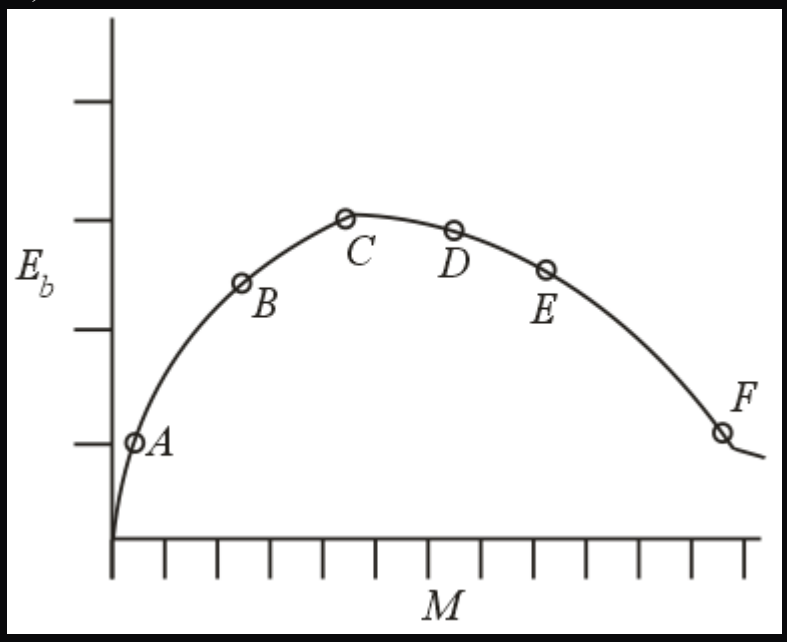
The above is a plot of binding energy per nucleon E_b, against the nuclear mass M; A, B, C, D, E, F correspond to different nuclei. Consider four reactions

i)$A + B \rightarrow C + \varepsilon$

ii)$C \rightarrow A + B + \varepsilon$

iii)$D + E \rightarrow F + \varepsilon$

iv)$F \rightarrow D + E + \varepsilon$



where ε is the energy released. In which reaction is ε positive?

(A) (i) and (iv)

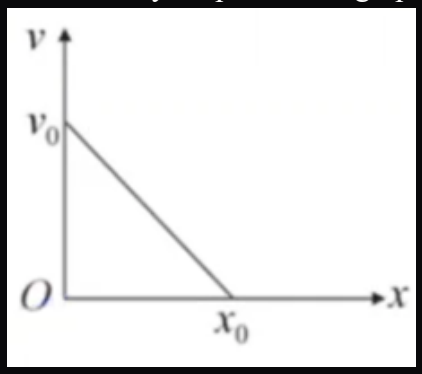
(B) (i) and (iii)

(C) (ii) and (iv)

(D) (ii) and (iii)

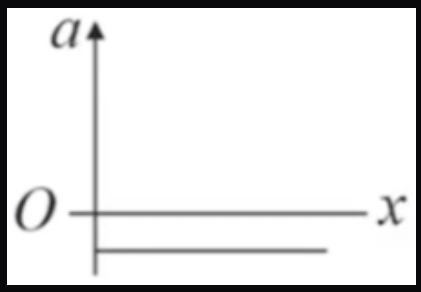
Q.8

The velocity-displacement graph ofa particle is shown in the figure.

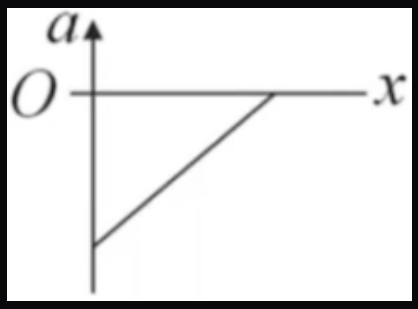


The acceleration-displacement graph of the same particle is represented by :

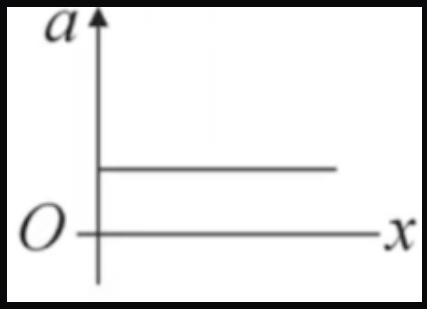
(A)



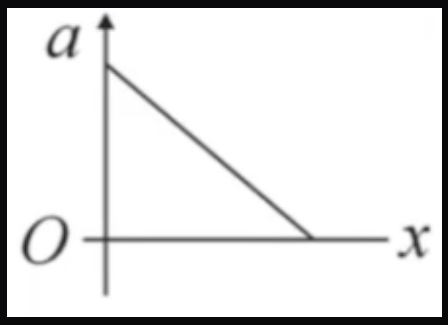
(B)



(C)



(D)



Q.9

The dimensions of magnetic field in M, L, T and C (coulomb) is given as

(A) $\left[\mathrm{MLT}^{-1}\mathrm{C}^{-1}\right]$

(B) $\left[\mathrm{MT}^{2}\mathrm{C}^{-2}\right]$

(C) $\left[\mathrm{MT}^{-1}\mathrm{C}^{-1}\right]$

(D) $\left[\mathrm{MT}^{-2}\mathrm{C}^{-1}\right]$

Q.10

The resonance frequency of LCR series a.c. circuit is f_0. Now the capacitance is made 4 times, then the new resonance frequency will become

(A) $\frac{f_0}{4}$

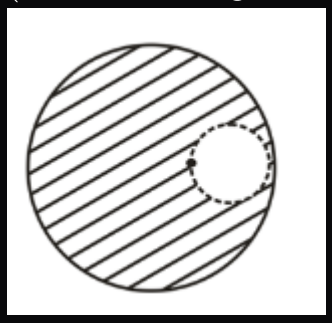
(B) $2f_0$

(C) f_0

(D) $\frac{f_0}{2}$

Q.11

From a solid sphere of mass M and radius R, a spherical portion of radius $\frac{\mathrm{R}}{2}$ is removed, as shown in the figure. Taking gravitational potential $\mathrm{V} = 0$ at $\mathrm{r} = \infty$ the potential at the centre of the cavity thus formed is $(\mathrm{G} = \text{Universal gravitational constant})$.



(A) $\frac{-2\mathrm{GM}}{3\mathrm{R}}$

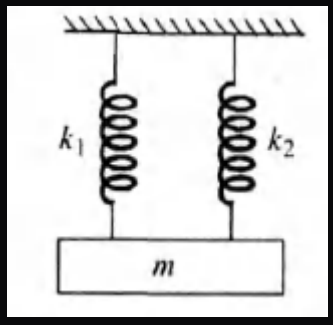
(B) $\frac{-2\mathrm{GM}}{\mathrm{R}}$

(C) $\frac{-\mathrm{GM}}{2\mathrm{R}}$

(D) $\frac{-\mathrm{GM}}{\mathrm{R}}$

Q.12

A mass m is suspended separately by two different springs in successive order, then time periods is t_1 and t_2 respectively. If m is connected by both springs as shown in figure, then time period is t_0, the correct relation is



(A) $t_0^2 = t_1^2 + t_2^2$

(B) $t_0^{-2} = t_1^{-2} + t_2^{-2}$

(C) $t_0^{-1} = t_1^{-1} + t_2^{-1}$

(D) $t_0 = t_1 + t_2$

Q.13

The circumference of the second orbit of an atom or ion having single electron, is 4×10^{-9} m. The de Broglie wavelength of electron revolving in this orbit should be:

(A) 2×10^{-9} m

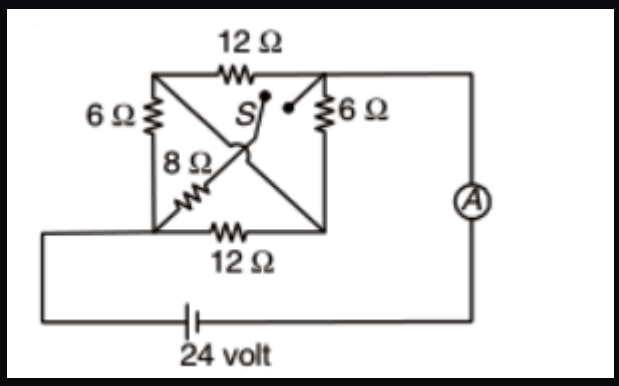
(B) 4×10^{-9} m

(C) 8×10^{-9} m

(D) 1×10^{-9} m

Q.14

In the circuit shown below, when the switch S is closed the reading of ammeter is changed by



(A) 3 A

(B) 2 A

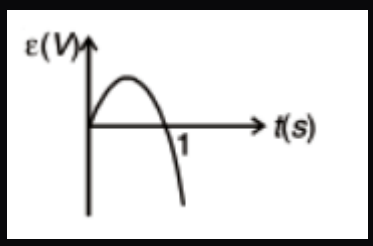
(C) 4 A

(D) 1 A

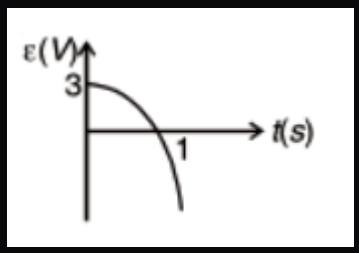
Q.15

The magnetic flux linked with a circuit is given by $\phi = t^3 - 3t - 7$. Choose the corrret graph between induced emf $[\varepsilon]$ and time $[f]$.

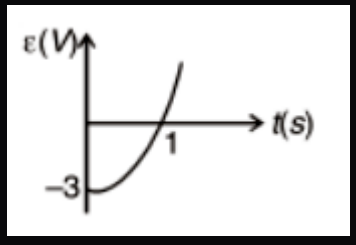
(A)



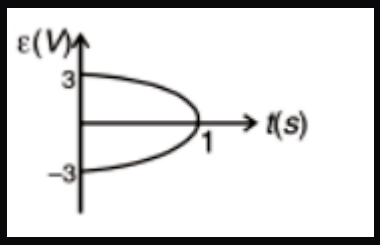
(B)



(C)



(D)



Q.16

The relation between Y, η and B is

(A) $\frac{1}{\eta} = \frac{1}{3\eta} + \frac{1}{9B}$

(B) $\frac{9}{Y} = \frac{1}{\eta} + \frac{3}{B}$

(C) $\frac{1}{\eta} = \frac{1}{B} + \frac{1}{Y}$

(D) $\frac{9}{Y} = \frac{3}{\eta} + \frac{1}{B}$

Q.17

Water falls from a height of $60\ \text{m}$ at the rate of $15\ \text{kg/s}$ to operate a turbine. The losses due to frictional force are 10% of energy. How much power is generated by the turbine $\left(g = 10\ \text{m/s}^2\right)$?

(A) $12.3\ \text{kW}$

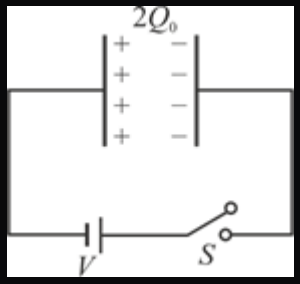
(B) $7.0\ \text{kW}$

(C) $8.1\ \text{kW}$

(D) $10.2\ \text{kW}$

Q.18

A capacitor of capacitance C having initial charge $2Q_0$, is connected to a battery of potential difference $V = \frac{Q_0}{C}$ as shown, then workdone by the battery is



(A) $\frac{Q_0^2}{2C}$

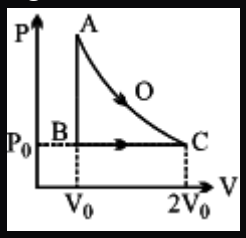
(B) $\frac{3Q_0^2}{2C}$

(C) $\frac{3Q_0^2}{C}$

(D) $\frac{2Q_0^2}{3C}$

Q.19

An ideal gas is taken from point A to point C on P-V diagram through two process AOC and ABC as shown in the figure. Process AOC is isothermal



(A) Process AOC requires more heat than process ABC.

(B) Process ABC requires more heat than process AOC.

(C) Both process AOC \& ABC require same amount of heat.

(D) Data is insufficient for comparison of heat requirement for the two processes.

Q.20

In a zener diode, the reverse bias voltage is 3 V and the width of the depletion region is 300 $\mathring{\mathrm{A}}$. The electric field intensity will be ___ $\mathrm{V\ cm^{-1}}$.

(A) 10^4

(B) 10^6

(C) 10^8

(D) 10^{-2}

Q.21

70 cal of heat is required to raise the temperature of 2 moles of an ideal gas from $30^{\circ}\mathrm{C}$ to $35^{\circ}\mathrm{C}$ while the pressure of the gas is kept constant. The amount of the heat required to raise the temperature of the same gas through the same temperature range at constant volume is (gas constant, $\mathrm{R} = 2\mathrm{calmol^{-1}\ K^{-1}}$) Mark answer in calories.

(A) Submit

Q.22

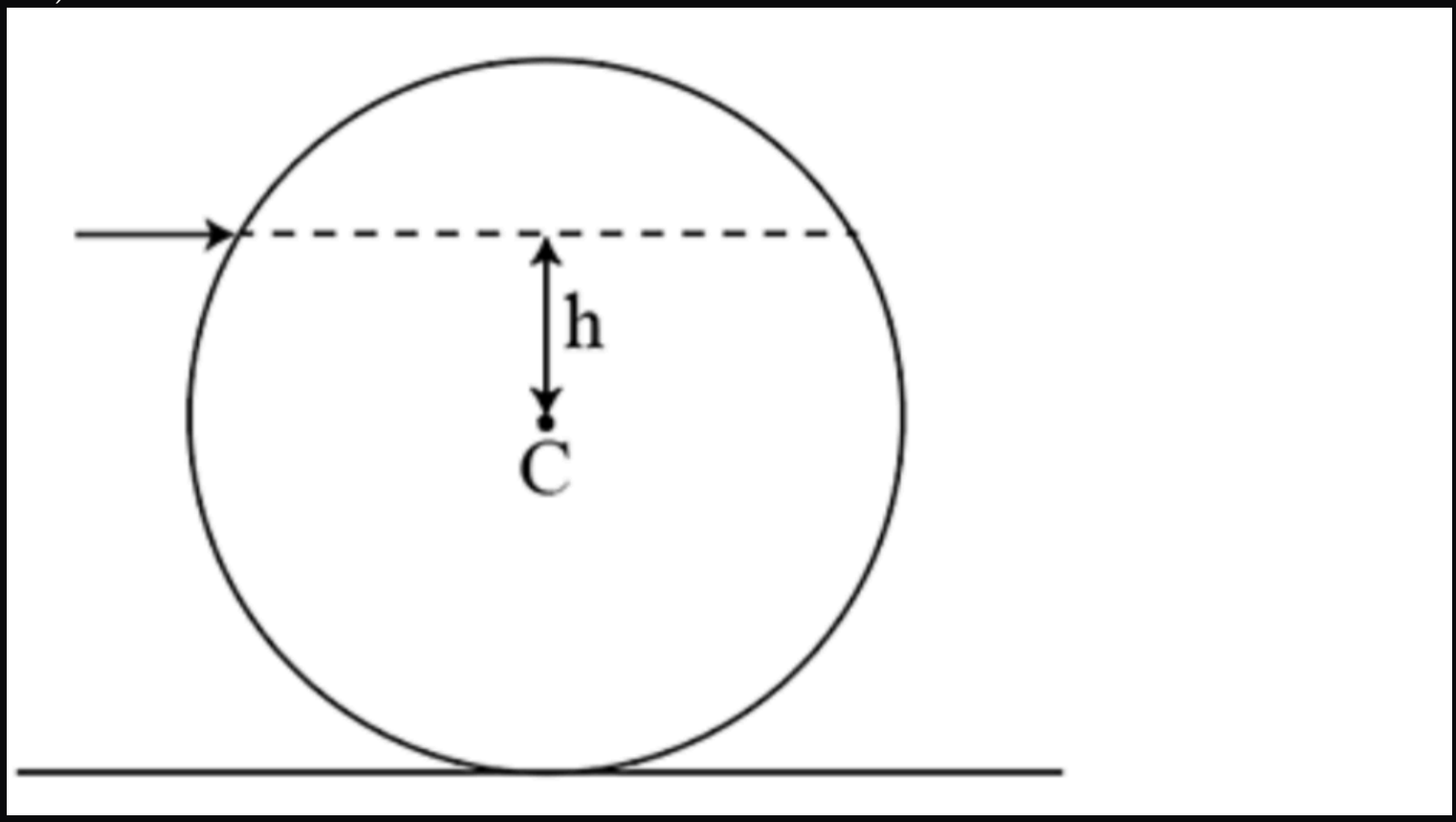
Two boys of masses $10\ \mathrm{kg}$ and $8\ \mathrm{kg}$ are moving along a vertical rope. The former climbing up with acceleration of $2\ \mathrm{ms^2}$ while later coming down with uniform velocity of $2\ \mathrm{ms^{-1}}$. Then tension in rope at fixed support will be (Take $g = 10\ \mathrm{ms^{-2}}$)

(A) Submit

Q.23

A billiard ball has mass $M = 250\ \mathrm{g}$ and radius $R = 2.5\ \mathrm{cm}$ and is initially at rest. A rod held horizontal at a height h above centre C hits the ball. The ball begins to roll without slipping. Find the value of h (mark answer in

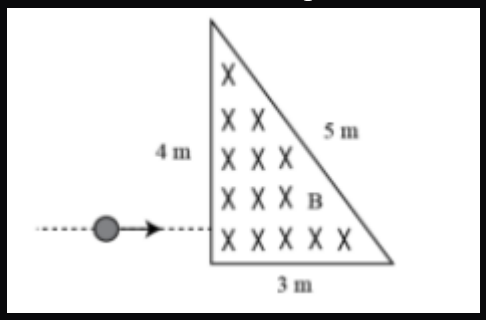
cm)



(A) Submit

Q.24

A small particle of mass $1\ \mathrm{kg}$ and charge $1\mathrm{C}$ enters perpendicularly in a triangular region of a uniform magnetic field of strength $2\ \mathrm{T}$ as shown in the figure. What is the maximum velocity (in ms^{-1}) of the particle with which it can enter so that it completes a half-circle in the region of the magnetic field?



(A) Submit

Q.25

The current flowing through a wire depends on time as $I = 3t^2 + 2t + 5$. The charge flowing the cross section of the wire in time from $t = 0$ to $t = 2\ \mathrm{s}$ is

(A) Submit

Q.26

Consider the reaction, $\mathrm{CaCO_3(\ s) \rightleftharpoons CaO(s) + CO_2(\ g)}$, in a closed container at equilibrium. At a fixed temperature, what will be the effect of adding more $\mathrm{CaCO_3}$ on the equilibrium concentration of $\mathrm{CO_2}$?

(A) It increases.

(B) It decreases.

(C) It remains the same.

(D) Cannot be predicated unless the value of K_P is known.

Q.27

Which of the following compounds is colored and paramagnetic?

(A) CuCl

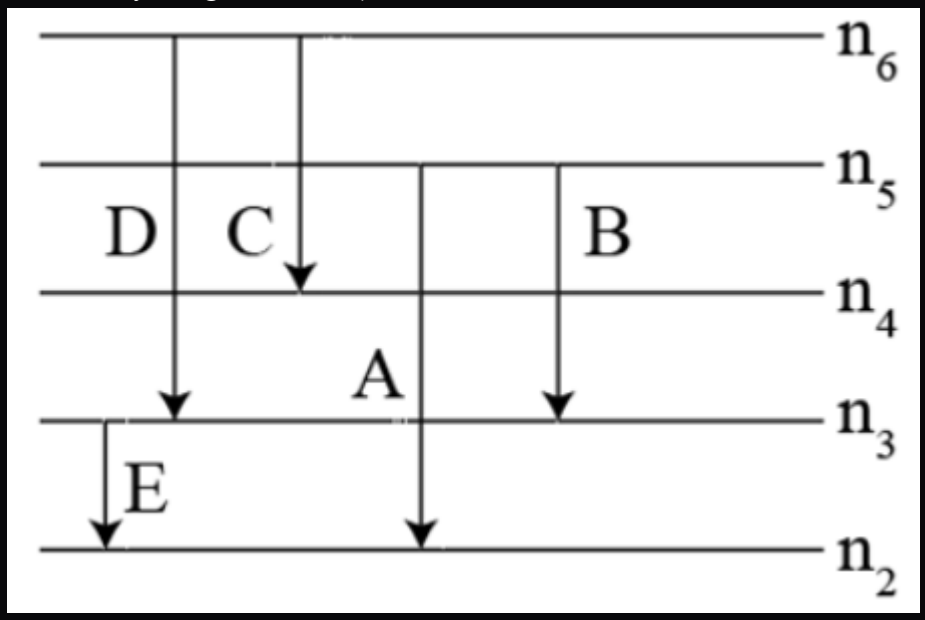
(B) $\mathrm{K_3[Cu(CN)_4]}$

(C) $\mathrm{CuF_2}$

(D) $\mathrm{[Cu(CH_3CN)_4]BF_4}$

Q.28

For a hypothetical H-like atom, which follows Bohr model, some spectral lines are observed as shown. If it is known that line E belong to visible region., then the line, which possibly belong to U.V. region is/are? $(\mathrm{n_1}$ is not necessarily the ground state)



(A) B and D

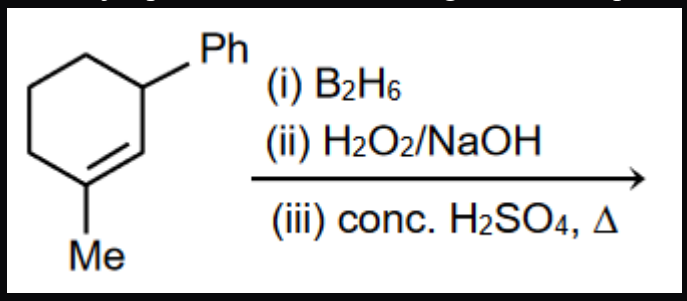
(B) D only

(C) C only

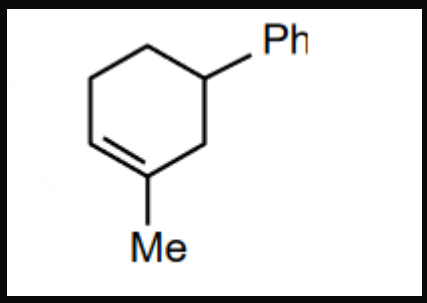
(D) A only

Q.29

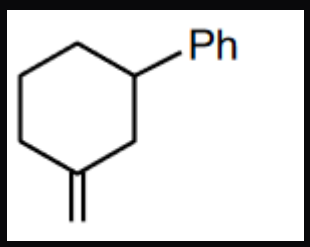
The major product of the following reaction sequence is



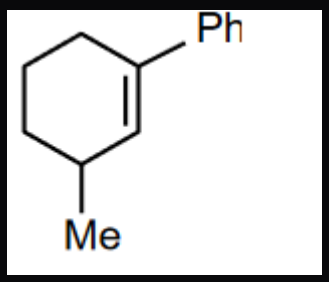
(A)



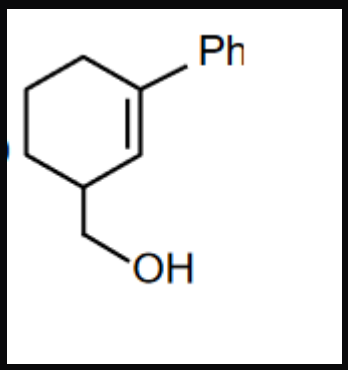
(B)



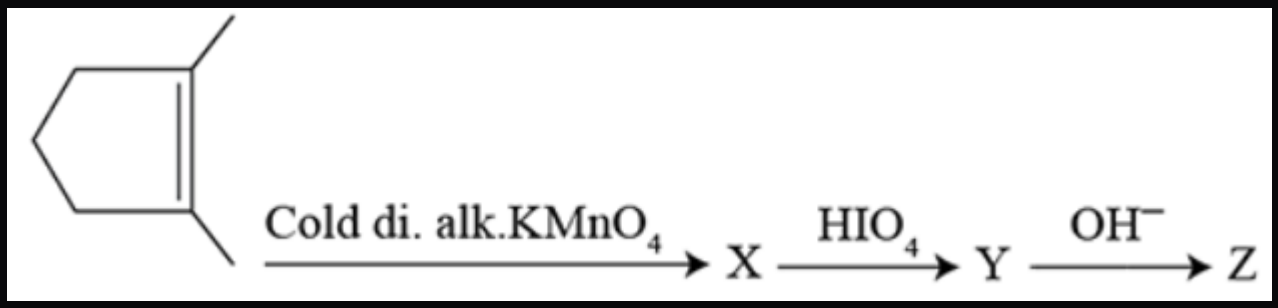
(C)



(D)

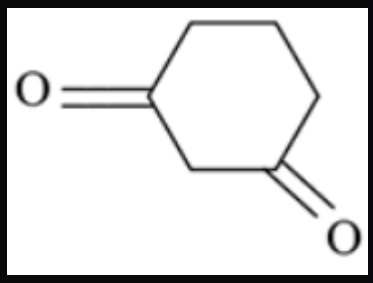


Q.30

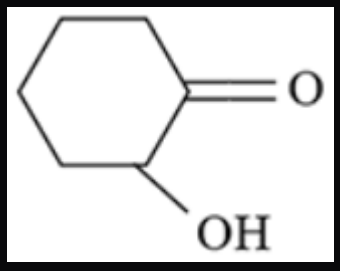


In the above sequence of reactions, Z is

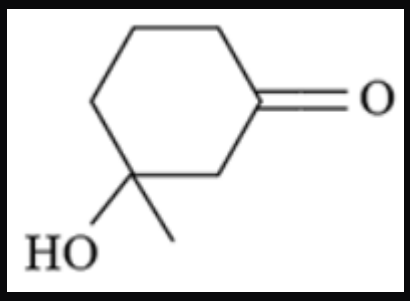
(A)



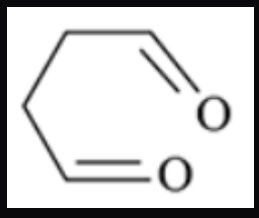
(B)



(C)



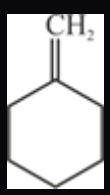
(D)



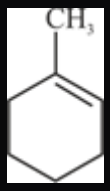
Q.31

In the reaction with HCl, an alkene reacts in accordance with the Markovnikov's rule, to give the product 1-chloro-1-methylcyclohexane. The possible alkene is

(A)

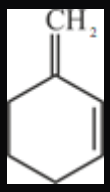


(B)



(C) Both (A) and (B)

(D)



Q.32

Which one is the most likely structure of $\mathrm{CrCl_3.6H_2O}$, if $1/3$ of total chlorine of the compound is precipitated by adding $\mathrm{AgNO_3}$ to its aqueous solution?

(A) $\mathrm{CrCl_3 \cdot 6H_2O}$

(B) $\mathrm{[Cr\,(H_2O)_3\,Cl_3] \cdot 3H_2O}$

(C) $\mathrm{[CrCl_2\,(H_2O)_4]\,Cl \cdot 2H_2O}$

(D) $\mathrm{[CrCl\,(H_2O)_5]\,Cl_2 \cdot H_2O}$

Q.33

$\mathrm{V_1\ mL}$ of NaOH of molarity X and $\mathrm{V_2\ mL}$ of $\mathrm{Ba(OH)_2}$ of molarity Y are mixed together. The mixture is completely neutralized by $\mathrm{100\ mL}$ of $\mathrm{0.1\ N\ HCl}$. If $\mathrm{V_1/V_2 = 1/4}$ and $\mathrm{X/Y = 4}$ then what fraction of acid is neutralized by $\mathrm{Ba(OH)_2}$

(A) 0.5

(B) 0.25

(C) 0.33

(D) 0.67

Q.34

Which of the following statements is/are INCORRECT ?

I. Synthesis of one molecule of glucose involves 18 molecules of ATP

II. Vitamin B 12 contains cobalt

III. Adenine and guanine are pyrimidines whereas thymine and cytosine are purines.

(A) I and II

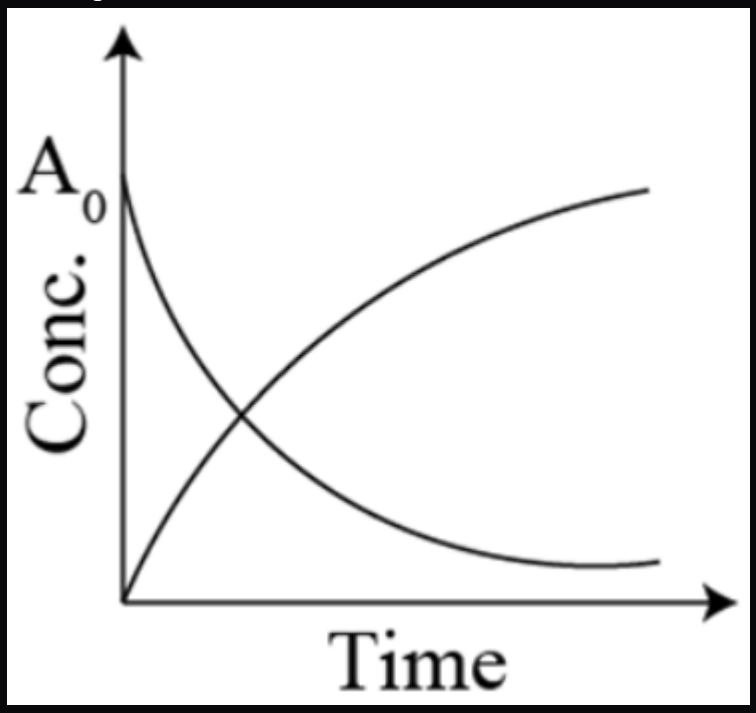
(B) Only III

(C) I, II and III

(D) None of these

Q.35

At the point of intersection of the two curves shown, the conc. of B is given by for, $\mathrm{A} \rightarrow \mathrm{nB}$



(A) $\frac{\mathrm{nA}_0}{2}$

(B) $\frac{A_0}{n-1}$

(C) $\frac{\mathrm{nA}_0}{\mathrm{n}+1}$

(D) $\left(\frac{\mathrm{n}-1}{\mathrm{n}+1}\right) \mathrm{A}_0$

Q.36

Given the electronic configurations of few elements. Select the incorrect Match:
\

Misplaced \h ine

\ \ \hline 2 & 1 s^2 2 s^2 2 p^3 &

An element belonging to 3rd period and 5th group

\ \ \hline 3 & 1 s^2 2 s^2 2 p^6 3 s^2 & \text { A d-block element } \ \ & 3 p^6 3 d^8 4 s^2 & \ \ \hline 4 & 1 s^2 2 s^2 2 p^6 3 s^2 3 p^6 & \text { An element from } 18^{\text {th }} \text { group } \ \ \hline \ \end{array} \

(A) 1 and 4 only

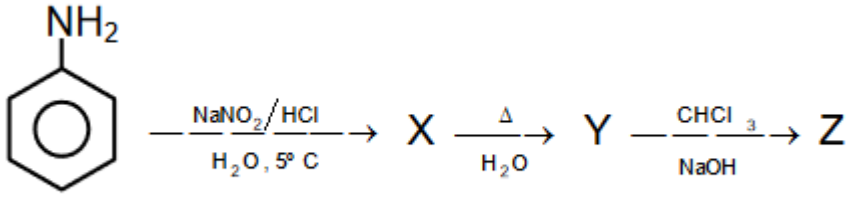
(B) 2 only

(C) 1, 2 and 3

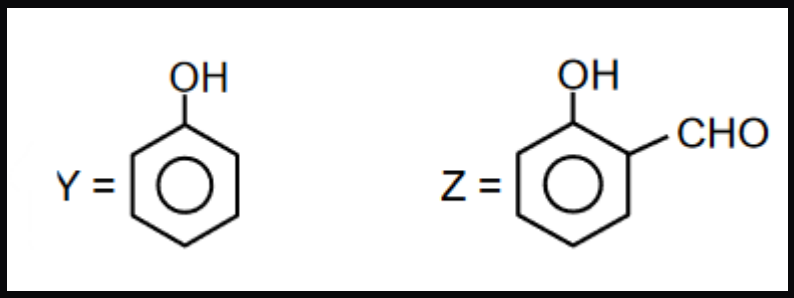
(D) All are the incorrect matches.

Q.37

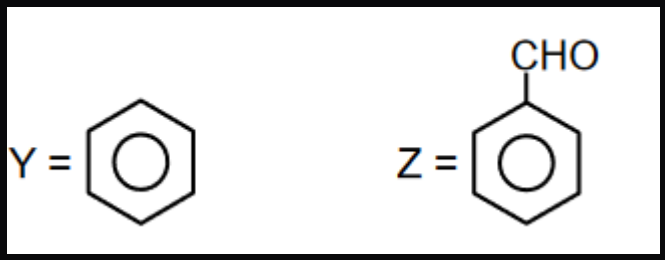
Identify the Y and Z in the following reaction.



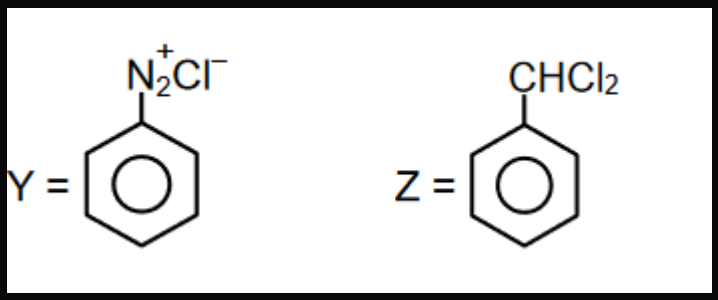
(A)



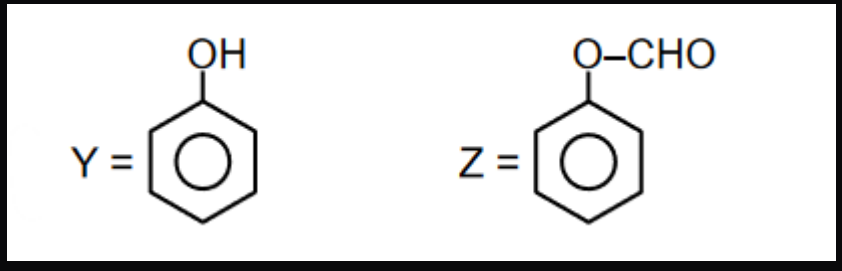
(B)



(C)



(D)



Q.38

The Henry's law constant for the solubility of N_2 gas in water is 1.0×10^5 atm. The mole fraction of N_2 in air is 0.8. The amount of N_2 from air dissolved in 10 mol of water at 298 K and 5 atm pressure is

(A) 4.0×10^{-4} mol

(B) 4.0×10^{-5} mol

(C) 5.0×10^{-4} mol

(D) 4.0×10^{-6} mol

Q.39

Assertion (A): The aqueous solution of CH_3COONa is alkaline in nature.
Reason (R): Acetate ion undergoes anionic hydrolysis.
The correct answer is:

(A) both (A) and (R) are true and (R) is the correct explanation of (A).

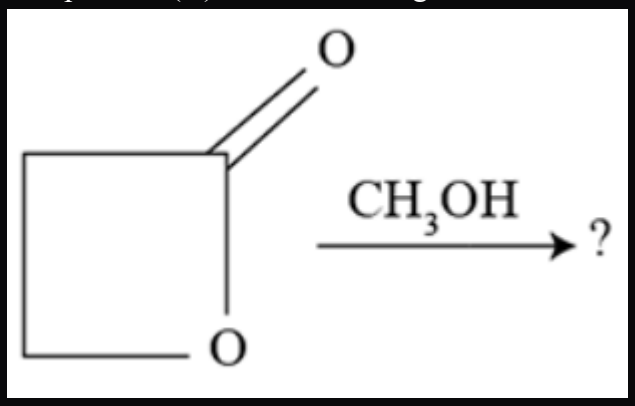
(B) both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is not true.

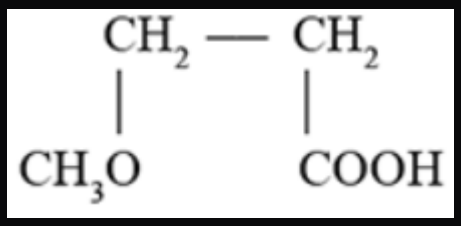
(D) (A) is not true but (R) is true.

Q.40

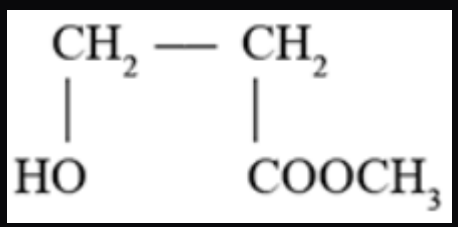
The product (A) of the following reaction



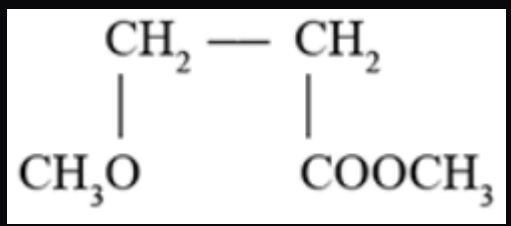
(A)



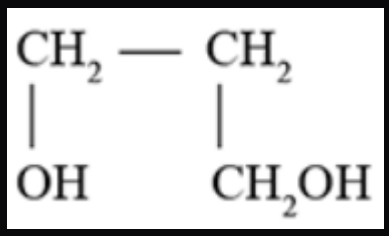
(B)



(C)



(D)



Q.41

The correct option(s) to distinguish nitrate salts of Mn^{2+} and Cu^{2+} taken separately is (are)
(1) Mn^{2+} shows the characteristic green colour in the flame test
(2) only Cu^{2+} shows the formation of precipitate by passing H_2S in acidic medium
(3) only Mn^{2+} shows the formation of precipitate by passing H_2S in faintly basic medium
(4) $Cu^{2+} \mid Cu$ has higher reduction potential than $Mn^{2+} \mid Mn$ (measured under similar conditions)

(A) 1,2

(B) 1,3

(C) 2,4

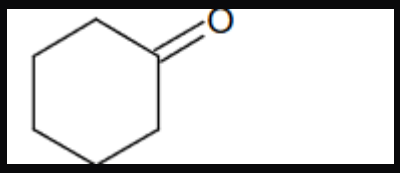
(D) $1, 2, 4$

Q.42

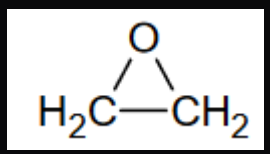
Which of the following compound produces primary alcohol on reaction with CH_3MgBr ?

(A) $CH_3 - CH = O$

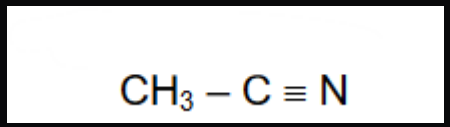
(B)



(C)



(D)



Q.43

Match the column I with column II and mark the appropriate choice.

Misplaced \h ine

(A) $(p) - (iii), (q) - (i), (r) - (ii), (s) - (iv)$

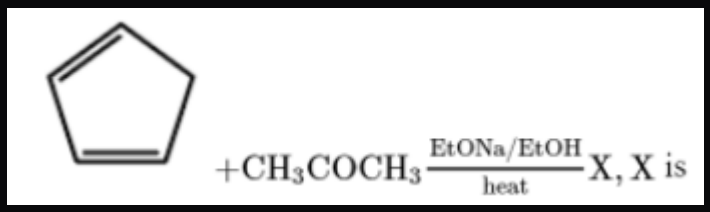
(B) $(p) - (iii), (q) - (iv), (r) - (i), (s) - (ii)$

(C) $(p) - (ii), (q) - (i), (r) - (iv), (s) - (iii)$

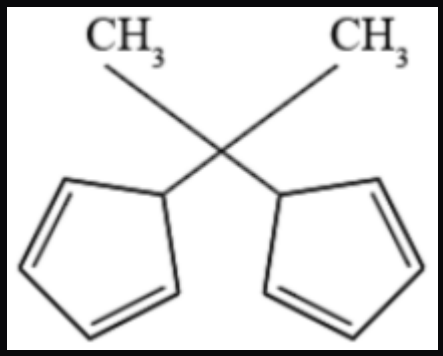
(D) (p) – (iv), (q) – (ii), (r) – (i), (s) – (iii)

Q.44

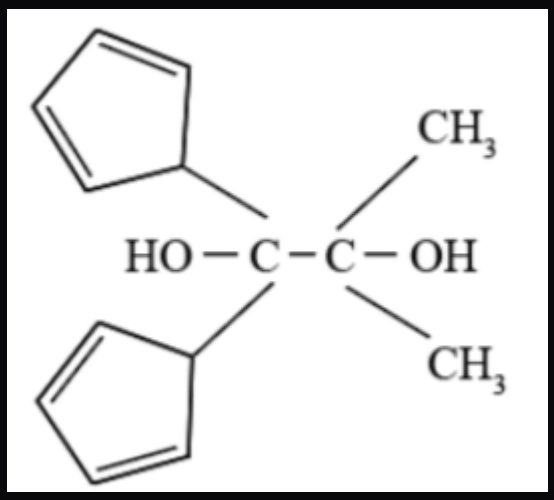
In the reaction



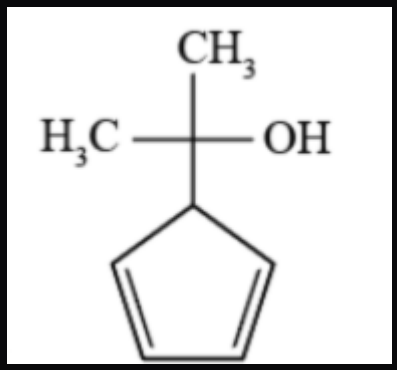
(A)



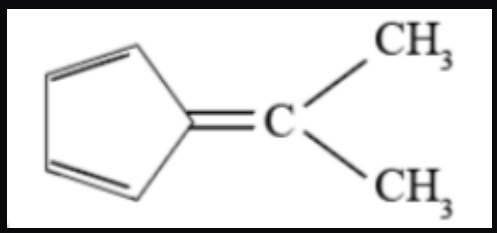
(B)



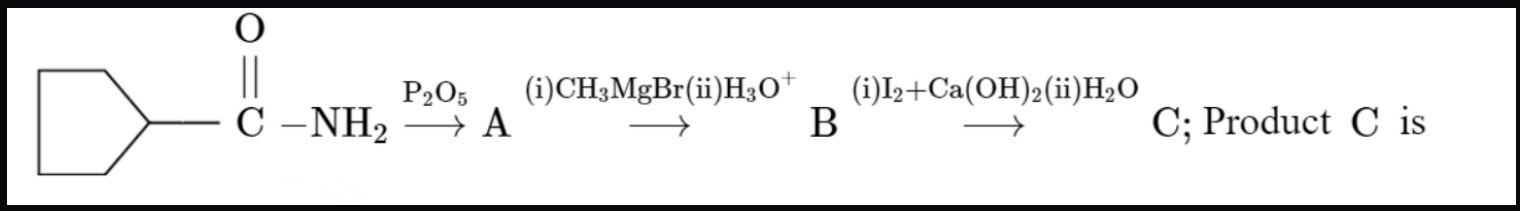
(C)



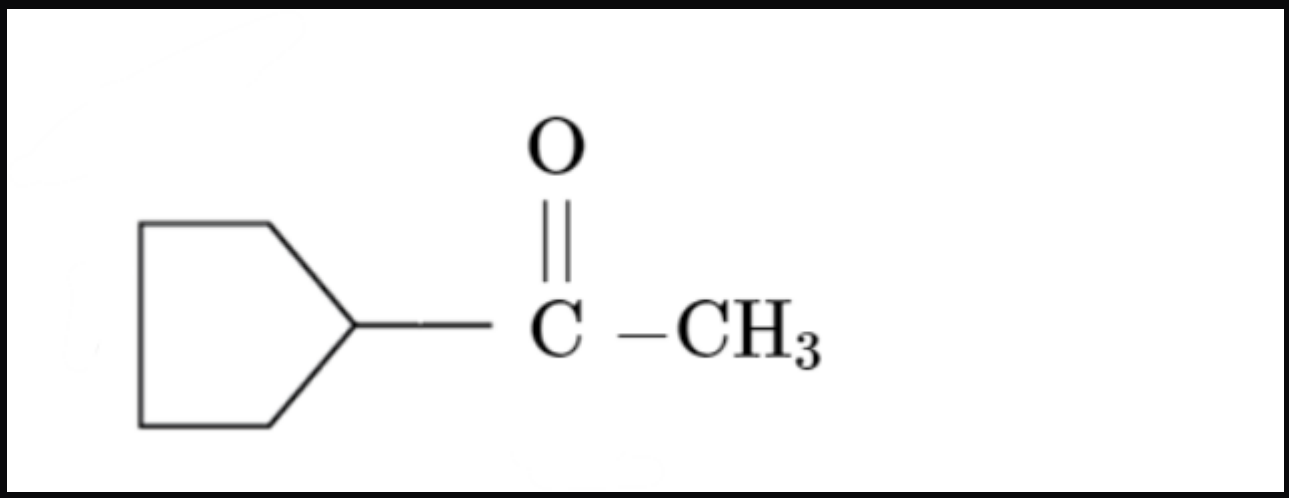
(D)



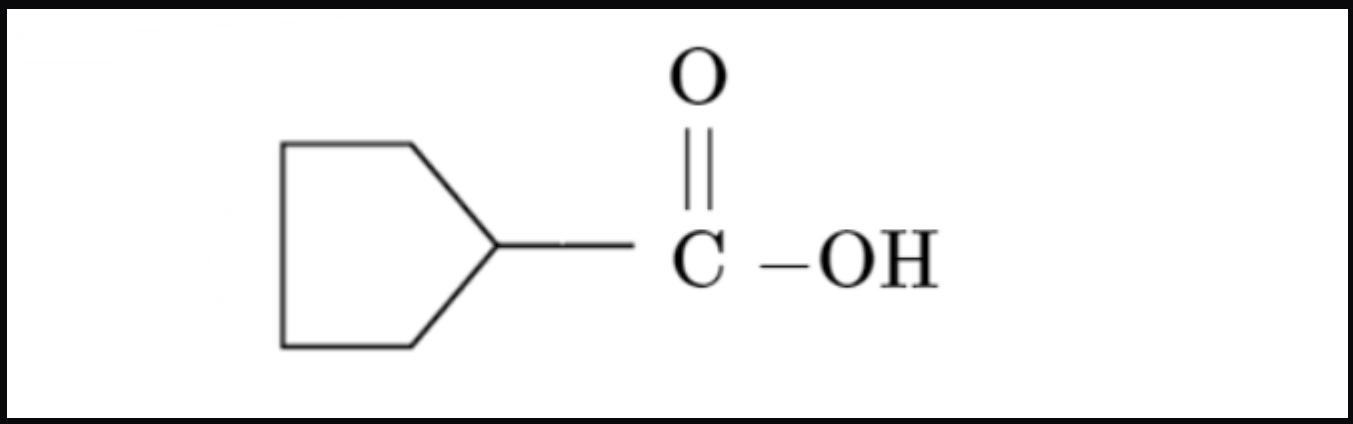
Q.45


O
C –NH2
P2O5
A
(i)CH3MgBr(ii)H3O+
B
(i)I2+Ca(OH)2(ii)H2O
C; Product C is

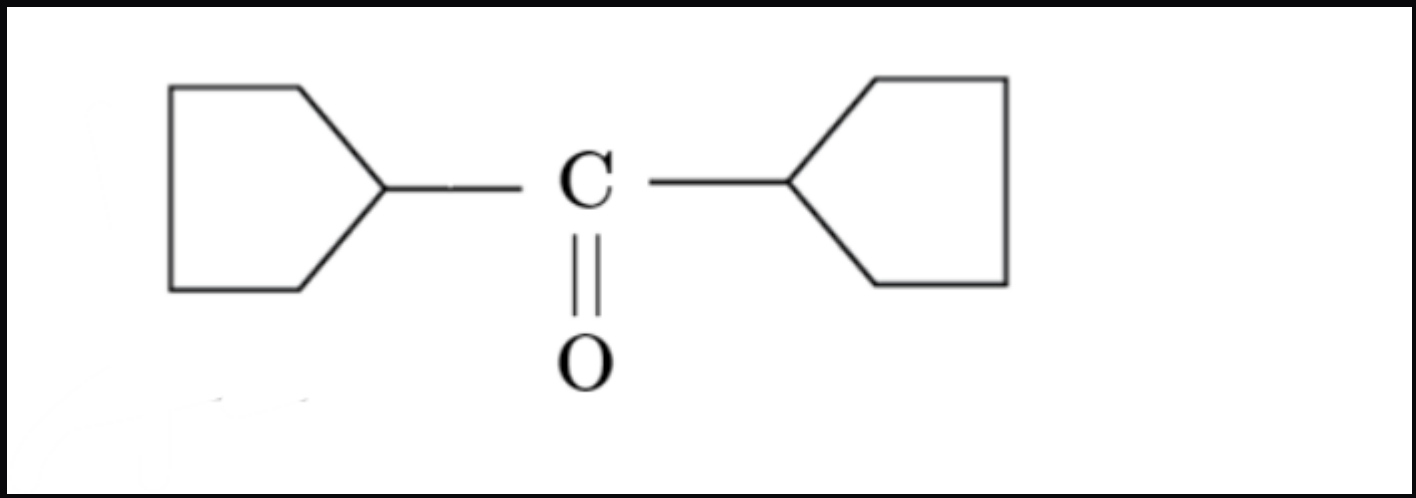
(A)


O
C –CH3

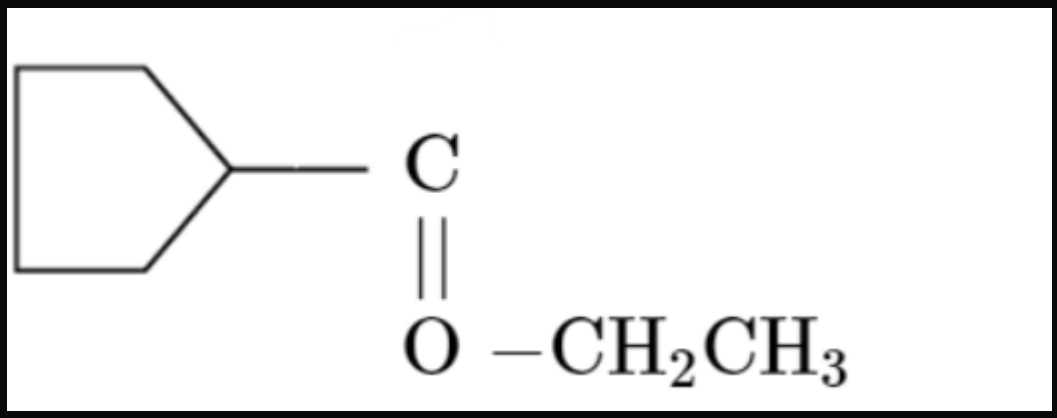
(B)


O
C –OH

(C)


C
O

(D)



Q.46

A current of 2 amp when passed for 5 hour through a molten salt deposits 22.2 g of metal of atomic mass 177. The oxidation state of the metal in the metal salt is
[Report your answer by rounding it upto nearest whole number]

(A) Submit

Q.47

The total number of cyclic structural as well as stereo isomers possible for a compound with the molecular formula C_5H_{10} is

(A) Submit

Q.48

In a neutral or faintly alkaline solution, 8 moles of permanganate anion quantitatively oxidize thiosulphate anions to produce X moles of a sulphur containing product. The magnitude of X is:

(A) Submit

Q.49

How many grams of glucose, $C_6H_{12}O_6$, should be dissolved in 0.5 kg of water at $25^\circ C$ to reduce the vapour pressure of water by 1%? (mark 10 times of mass of glucose as answer)

(A) Submit

Q.50

The work function (ϕ) of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is :

Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
ϕ (eV)	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

(A) Submit

Q.51

The area (in sq. units) bounded by $x^2 + y^2 = 1$ and the curve $y^2 \geq x^2$, above the x-axis is

(A) $\frac{1}{4}$

(B) $\frac{\pi}{4}$

(C) $\frac{1}{6}$

(D) $\frac{\pi}{6}$

Q.52

Statement I: The point $A(1, 0, 7)$ is the mirror image of the point $B(1, 6, 3)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$
Statement II: The line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ bisects the line segment joining $A(1, 0, 7)$ and $B(1, 6, 3)$.

(A) Statement I is true, statement II is true; statement II is not a correct explanation for statement I

(B) Statement I is true, statement II is false

(C) Statement I is false, statement II is true

(D) Statement I is true, statement II is true; statement II is a correct explanation for statement I

Q.53

If A and B are square matrices of order 3 such that $|A| = 3$ and $|B| = 2$, then the value of $\left|A^{-1}adjB^{-1}adj\left(3A^{-1}\right)\right|$ is equal to

(A) 27

(B) $\frac{27}{4}$

(C) $\frac{1}{108}$

(D) $\frac{1}{4}$

Q.54

The solution of the differential equation $\frac{dy}{dx} = \frac{y^2 + x\ln x}{2xy}$ is (where, c is the constant of integration)

(A) $2x^2 = y(\ln x)^2 + 2cy$

(B) $2y^2 = x(\ln x)^2 + 2cx$

(C) $x^2 = y(\ln x)^2 + c$

(D) $2y^2 = \frac{x}{y}(\ln x)^2 + cx$

Q.55

The number of five-digit numbers formed with the digits $0, 1, 2, 3, 4$ and 5 (without repetition) and divisible by 6 are

(A) 72

(B) 84

(C) 96

(D) 108

Q.56

The equation of the image of line $y = x$ with respect to the line mirror $2x - y = 1$ is

(A) $y = 7x - 5$

(B) $y = 7x - 6$

(C) $y = 3x - 7$

(D) $y = 6x - 5$

Q.57

Let $f : R \to R$ be a function defined by $f(x) = -x^3 - 3x^2 - 6x + 1$. Number of integers in the solution set of x satisfying the inequality $f\big(f\big(x^3 + f(x)\big)\big) \geq f\big(f\big(-f(x) - x^3\big)\big)$ is

(A) 3

(B) 4

(C) 5

(D) 6

Q.58

In the ΔOAB, M is the midpoint of AB, C is a point on OM, such that $2\overrightarrow{OC} = \overrightarrow{CM}$. X is a point on the side OB, such that $\overrightarrow{OX} = 2\overrightarrow{XB}$. The line XC is produced to meet OA at Y, then $\frac{OY}{YA}$ is equal to

(A) $\frac{1}{3}$

(B) $\frac{2}{7}$

(C) $\frac{3}{2}$

(D) $\frac{2}{5}$

Q.59

A and B are two independent witnesses (i.e. there is no collusion between them) in a case. The probability that A will speak the truth is x and the probability that B will speak the truth is y. A and B agree in a certain statement. The probability that the statement is true is

(A) $\frac{x-y}{x+y}$

(B) $\frac{xy}{1+x+y+xy}$

(C) $\frac{x-y}{1-x-y+2xy}$

(D) $\frac{xy}{1-x-y+2xy}$

Q.60

If only the 4th term in the expansion of $\left(2+\frac{3x}{8}\right)^{10}$ has the greatest numerical value, then the integral values of x are

(A) $\{-3,-2,2,3\}$

(B) $\{-2,-1,1,2\}$

(C) $\{-3,3\}$

(D) $\{-3,-2,-1,0,1,2,3\}$

Q.61

Let $f:R\to R$ be a function such that $f\left(\frac{x+y}{3}\right)=\frac{f(x)+f(y)}{3}$, $f(0)=0$ and $f'(0)=5$, then

(A) $f(x)$ is a quadratic function

(B) $f(x)$ is continuous but not differentiable

(C) $f(x)$ is differentiable in R

(D) $f(x)$ is bounded in R

Q.62

If the range of $f(x)=\tan^{-1}x+2\sin^{-1}x+\cos^{-1}x$ is $[a,b]$, then

(A) $a=\frac{\pi}{4}$

(B) $a=-\frac{\pi}{2}$

(C) $b=\frac{5\pi}{4}$

(D) $b=\frac{3\pi}{2}$

Q.63

The discriminant of the quadratic
$(2x+1)^2+(3x+2)^2+(4x+3)^2+\ldots n$ terms $=0, \forall n>3, x\in R$ is

(A) positive

(B) zero

(C) negative

(D) depends on n

Q.64

If a circle of radius R passes through the origin O and intersects the coordinate axes at A and B, then the locus of the foot of perpendicular from O on AB is:

(A) $\left(x^2+y^2\right)(x+y)=R^2xy$

(B) $\left(x^2+y^2\right)^3=4R^2x^2y^2$

(C) $\left(x^2+y^2\right)^2=4R^2x^2y^2$

(D) $\left(x^2+y^2\right)^2=4Rx^2y^2$

Q.65

Let $A=\left[a_{ij}\right]_{3\times3}$ be a square matrix such that $AA^T=4I, |A|<0$. If $\begin{vmatrix} a_{11}+4 & a_{12} & a_{13} \\ a_{21} & a_{22}+4 & a_{23} \\ a_{31} & a_{32} & a_{33}+4 \end{vmatrix}=5\lambda|A+I|$. Then, λ is equal to

(A) $\frac{4}{5}$

(B) $-\frac{4}{5}$

(C) $\frac{8}{5}$

(D) $-\frac{8}{5}$

Q.66

Consider $\mathrm{I}_1=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\frac{e^{\sin x}+1}{e^{\cos x}+1}dx$ and $\mathrm{I}_2=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\frac{e^{\cos x}+1}{e^{\sin x}+1}dx$, then

(A) $\mathrm{I}_1>\mathrm{I}_2$

(B) $\mathrm{I}_1<\mathrm{I}_2$

(C) $\mathrm{I}_1=\mathrm{I}_2$

(D) $\mathrm{I}_1+\mathrm{I}_2=0$

Q.67

The number of values of x such that $x, [x]$ and $\{x\}$ are in arithmetic progression is equal to (where, $[\cdot]$ denotes the greatest integer function and $\{\cdot\}$ denotes the fractional part function)

(A) 0

(B) 1

(C) 2

(D) 4

Q.68

The real part of the complex number z satisfying $|z-1-2i|\leq1$ and having the least positive argument, is

(A) $\frac{4}{5}$

(B) $\frac{8}{5}$

(C) $\frac{6}{5}$

(D) $\frac{7}{5}$

Q.69

For two data sets, each with size 5, the variances are given to be 3 and 4 and the corresponding means are given to be 2 and 4, respectively. The variance of the combined data set is

(A) $\frac{11}{2}$

(B) $\frac{9}{2}$

(C) $\frac{13}{2}$

(D) $\frac{5}{2}$

Q.70

Let $A = \{1,\ 2,\ 3,\ 4,\ 5\}$ and $f:\ A \to A$ be an injective function such that $f(i) \neq i,\ \forall\, i \in\ A$ then number of such functions f are:

(A) 120

(B) 96

(C) 44

(D) None of these

Q.71

Let $f(x) = \tan^3 x - 3\tan^2 x + 1, \forall x \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$. Then, the positive difference between the least value and the local maximum of the function is

(A) Submit

Q.72

Sanjay has 10 friends among whom two are married to each other. He wishes to invite 5 of them for a party. If the married couple refuse to attend separately, then the number of different ways in which he can invite five friends is

(A) Submit

Q.73

For $x\varepsilon\mathbf{R}$, let $[x]$ denote the greatest integer $\leq x$. Largest natural number n for which $E = \left[\frac{\pi}{2}\right] + \left[\frac{1}{100} + \frac{\pi}{2}\right] + \left[\frac{2}{100} + \frac{\pi}{2}\right] \ldots + \left[\frac{n}{100} + \frac{\pi}{2}\right] < 43$, is

(A) Submit

Q.74

If $I_n = \int_0^{n\pi} \max\left(|\sin x|, \left|\sin^{-1}(\sin x)\right|\right) dx$, then $I_2 + I_4$ has the value $\frac{\lambda\pi^2}{2}$; where λ is

(A) Submit

Q.75

Let $\Delta(x)=\begin{vmatrix} 3+2\sin^4 x & 2\cos^4 x & \sin^2 2x \\ 2\sin^4 x & 3+2\cos^4 x & \sin^2 2x \\ 2\sin^4 x & 2\cos^4 x & 3+\sin^2 2x \end{vmatrix}$ then $\int_{-\pi/2}^{\pi/2} x\Delta(x)dx$ equals

(A) Submit

Answers & Solutions

Q.1 Answer:

$R=\frac{2R_1R_2}{R_1+R_2}$

Solution:

For upward projection $\mathrm{R_{max}}=\frac{\mathrm{v_0^2}}{\mathrm{g(1+\sin\theta)}}=\mathrm{R_1}$

for downward projection $R_{\max}=\frac{v_0^2}{g(1-\sin\theta)}=R_2$

for projection on ground $R_{\max}=\frac{v_0^2}{g}=R\therefore\frac{2}{\mathrm{R}}=\frac{1}{\mathrm{R_1}}+\frac{1}{\mathrm{R_2}}\Rightarrow\mathrm{R}=\frac{2\mathrm{R_1}R_2}{\mathrm{R_1}+\mathrm{R_2}}$

Q.2 Answer:

zero

Solution:

The mass m is not moving with respect to the lift and also has no tendency to move. Hence, friction force acting on it is equal to zero.

Q.3 Answer:

Assertion is false but, reason is true.

Solution:

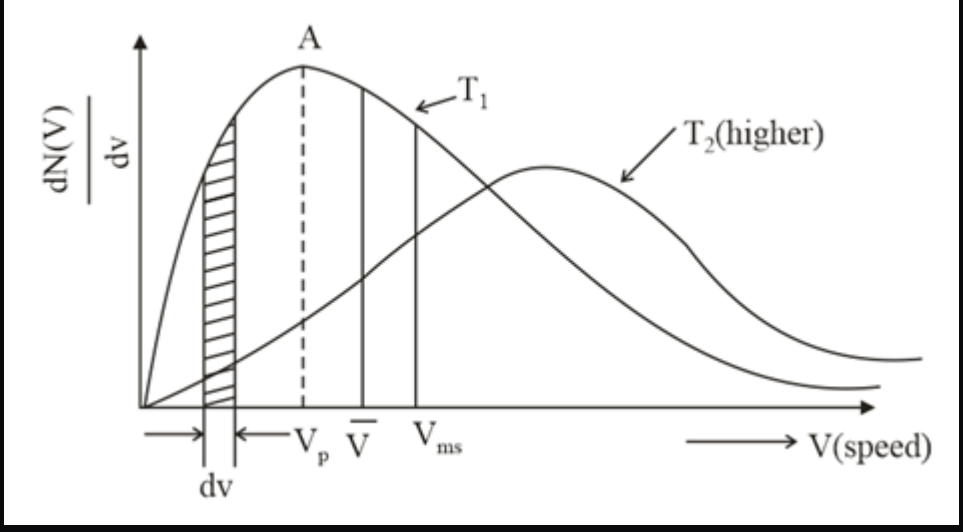
Quality (Wave form) of sound distinguish the different sources of sound from each other
So, assertion is false but, reason is true.

Q.4 Answer:

If both assertion and reason are true and the reason is correct explanation of the assertion.

Solution:

From the Maxwellian distribution curve,



v_p =**Most probable speed**

$\overline{v}$ =**Average or mean speed**

v_{rms} =**Root mean square speed**

Q.5 Answer:

$\sqrt{\frac{\mathrm{Kq}}{2\mathrm{mR}}\left(\frac{\mathrm{q}}{2}+2\mathrm{Q}\right)}$

Solution:

$$\Delta \mathrm{KE}+\Delta \mathrm{PE}=0$$
$$2\cdot\left[\frac{1}{2}\mathrm{mv}^2\right]+\left[0-\left(\frac{\mathrm{Kq}^2}{4\mathrm{R}}+\frac{2\mathrm{KqQ}}{2\mathrm{R}}\right)\right]=0$$
$$\mathrm{v}=\sqrt{\frac{1}{\mathrm{m}}\left(\frac{\mathrm{Kq}^2}{4\mathrm{R}}+\frac{\mathrm{KqQ}}{\mathrm{R}}\right)}$$

Q.6 Answer:

$3\ \mathrm{cm}$

Solution:

Min. width of plane mirror to see full face $=\frac{D-d}{2}=\frac{10-4}{2}=3\ \mathrm{cm}$

Q.7 Answer:

(i) and (iv)

Solution:

Energy ε is related only when lighter nuclei fuse to form a heavier nucleus such as in reaction (i)

$$A+B\rightarrow C+\varepsilon$$

Again ,energy is released when a heavy nucleus splits into lighter nuclei as in(iv)

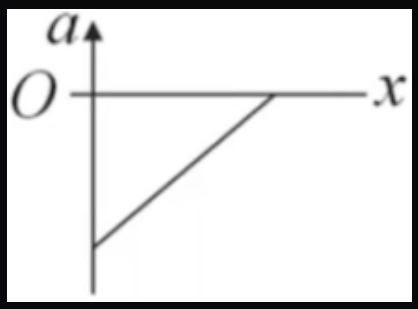
$$F\rightarrow D+E+\varepsilon$$

But the remaining cases it is not possible that lighter nuclei fuse to form a heavier nucleus and a heavy nucleus splits into lighter nuclei.

that cases are given below

$D+E\rightarrow F+\varepsilon$ *and* $C\rightarrow A+B+\varepsilon$

Q.8 Answer:



Solution:

Equation of $v = -\left(\frac{v_0}{x_0}\right)x + v_0 \rightarrow$ **(1)**

Differentiating both sides

$$\frac{dv}{dt} = -\left(\frac{v_0}{x_0}\right)\frac{dx}{dt}$$

$$\Rightarrow a = -\left(\frac{v_0}{x_0}\right)(v) \quad \left[v = \frac{dx}{dt}\right]$$

$$\Rightarrow a = -\frac{v_0}{x_0}\left[-\left(\frac{v_0}{x_0}\right)x + v_0\right] \quad [\text{from equation 1}]$$

$$\Rightarrow a = \left(\frac{v_0}{x_0}\right)^2 x - \frac{v_0^2}{x_0}$$

[from equation 1

$\Rightarrow a - x$ **graph will be a straight line with positive slope**

$\left[m = \left(\frac{v_0}{x_0}\right)^2\right]$ and-ve y-intercept $\left[C = -\frac{v_0^2}{x_0}\right]$

Q.9 Answer:

$[MT^{-1}C^{-1}]$

Solution:

Force $F = qvB$

$$[MLT^{-2}] = [C][LT^{-1}][B]$$

$$\Rightarrow [B] = [MC^{-1}T^{-1}]$$

Q.10 Answer:

$\frac{f_0}{2}$

Solution:

In LCR series circuit, resonance frequency f_0 is given by

$$L\omega = \frac{1}{C\omega} \Rightarrow \omega^2 = \frac{1}{LC} \quad \therefore \omega = \sqrt{\frac{1}{LC}} = 2\pi f_0$$

$$\therefore f_0 = \frac{1}{2\pi\sqrt{LC}} \quad \text{or } f_0 \propto \frac{1}{\sqrt{C}}$$

When the capacitance of the circuit is made 4 times, its resonant frequency become f_0'

$$\therefore \frac{f_0'}{f_0} = \frac{\sqrt{C}}{\sqrt{4C}} \text{ or } f_0' = \frac{f_0}{2}$$

Q.11 Answer:

$\frac{-\mathrm{GM}}{\mathrm{R}}$

Solution:

Potential at the centre of cavity before removing the spherical portion,

$\mathrm{V}_1 = \frac{-\mathrm{GM}}{2\mathrm{R}^3}\left(3\mathrm{R}^2 - \left(\frac{\mathrm{R}}{2}\right)^2\right) = \frac{-\mathrm{GM}}{2\mathrm{R}^3}\left(3\mathrm{R}^2 - \frac{\mathrm{R}}{4}^2\right) = \frac{-11\mathrm{GM}}{8\mathrm{R}}$

Mass of spherical portion to be removed,

$\mathrm{M'} = \frac{\mathrm{MV'}}{\mathrm{V}} = \frac{\mathrm{M}\frac{4\pi}{3}\left(\frac{\mathrm{R}}{2}\right)^3}{\frac{4\pi}{3}\mathrm{R}^3} = \frac{\mathrm{M}\frac{\mathrm{R}^3}{8}}{\mathrm{R}^3} = \frac{\mathrm{M}}{8}$

Potential at the centre of cavity due to spherical portion to be removed

$\mathrm{V}_2 = \frac{-3\mathrm{GM'}}{2\mathrm{R'}} = \frac{-3\mathrm{G}(\mathrm{M}/8)}{2(\mathrm{R}/2)} = \frac{-3\mathrm{GM}}{8\mathrm{R}}$

Potential at the centre of cavity formed

$\mathrm{V_P} = \mathrm{V}_1 - \mathrm{V}_2 = \frac{-11\mathrm{GM}}{8\mathrm{R}} - \left(\frac{-3\mathrm{GM}}{8\mathrm{R}}\right) = \frac{-\mathrm{GM}}{\mathrm{R}}$

Q.12 Answer:

$t_0^{-2} = t_1^{-2} + t_2^{-2}$

Solution:

As, $t_1 = 2\pi\sqrt{\frac{m}{k_1}}$

or $t_1^2 = \frac{2\pi^2 m}{k_1}$

or $k_1 = \frac{4\pi^2 m}{t_1^2}$

Similarly, $k_2 = \frac{4\pi^2 m}{t_2^2}$

and $(k_1 + k_2) = \frac{4\pi^2 m}{t_0^2}$

$\therefore \frac{4\pi^2 m}{t_0^2} = \frac{4\pi^2 m}{t_1^2} + \frac{4\pi^2 m}{t_2^2}$

or $\frac{1}{t_0^2} = \frac{1}{t_1^2} + \frac{1}{t_2^2}$

Q.13 Answer:

$2 \times 10^{-9}\ \mathrm{m}$

Solution:

We know that, angular momentum,

$mvr = \frac{nh}{2\pi} \Rightarrow 2\pi\mathrm{r} = \mathrm{n}\left(\frac{\mathrm{h}}{\mathrm{mv}}\right)$ According to de broglie's wavelength, $\lambda = \frac{\mathrm{h}}{\mathrm{mv}}$

$\Rightarrow 2\pi r = n\lambda$

Where r is the radius of the orbit, λ is the de Broglie wavelength and n represents integral quantisation for standing wave(electron)

$\therefore \lambda = \frac{2\pi r}{n} = \frac{4\times10^{-9}}{2} = 2 \times 10^{-9}\ \text{m}$

Q.14 Answer:

3 A

Solution:

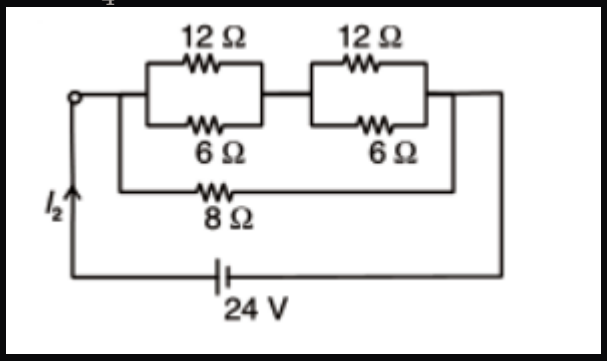
When switch is open,

$I_1 = \frac{V}{R_{eq}} = \frac{24}{8}$

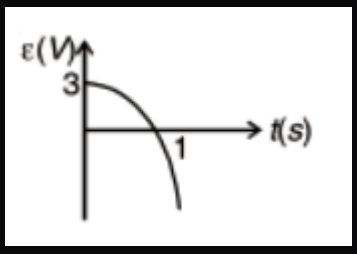
$I_1 = 3\ \text{A}$

When switch S is closed,

$I_2 = \frac{24}{4} = 6\ \text{A}$



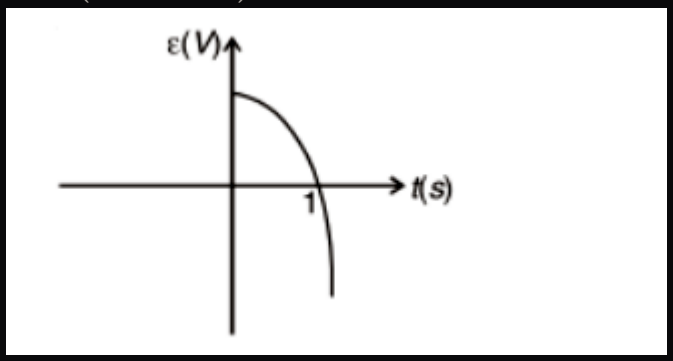
Q.15 Answer:



Solution:

$\varepsilon = \frac{-d\phi}{dt} = -\left(3t^2 - 3\right) V$

$\varepsilon = (-3t^2 + 3)V$



Q.16 Answer:

$\frac{9}{Y} = \frac{3}{\eta} + \frac{1}{B}$

Solution:

The relation between Y, η and B is $\frac{9}{Y} = \frac{1}{B} + \frac{3}{\eta}$

where Y is Young's modulus, B is Bulk modulus and η modulus of rigidity.

Q.17 Answer:

$8.1\ \text{kW}$

Solution:

Power given to turbine $= \frac{mgh}{t}$

$P_{\text{in}} = \left(\frac{m}{t}\right) gh = P_{\text{in}} = 15 \times 10 \times 60$

$P_{\text{in}} = 9000\ \text{W}$

$\Rightarrow P_{\text{in}} = 9\ \text{kW}$

As efficiency of turbine is 90% therefore power generation

$= 90\%$ of $9\ \text{kW}$

$\Rightarrow P_{\text{out}} = 9 \times \frac{90}{100}$

$\Rightarrow P_{\text{out}} = 8.1\ \text{kW}$

Q.18 Answer:

$\frac{3Q_0^2}{C}$

Solution:

Charge through the battery

$= Q_0 - (-2Q_0) = 3Q_0$

Work done by battery

$= \frac{(3Q_0)Q_0}{C} = \frac{3Q_0^2}{C}$

Q.19 Answer:

Process AOC requires more heat than process ABC.

Solution:

$\text{T}_\text{A} = \text{T}_\text{C}$

Hence $\text{dU} = 0$ for all processes from A to C

$\text{dQ} = \text{dW} =$ area under graph

which is maximum along curve AOC in given figure

Q.20 Answer:

10^6

Solution:

Given, reverse bias voltage of zener diode, $V = 3\ \text{V}$ and the width of the depletion region, $d = 300\ \text{Å}$.

The electric field intensity at depletion region is given by,

$E = \frac{V}{d}$

$E = \frac{3}{300 \times 10^{-10}}\ \mathrm{V\ m^{-1}}$

$E = 10^6\ \mathrm{V\ cm^{-1}}$ **will be electric field at depletion region.**

Q.21 Answer:

—

Solution:

$$\mathrm{C_p} = \frac{\mathrm{Q}}{\mathrm{n}\Delta\theta} = \frac{70}{2 \times 5} = 7\mathrm{calmol^{-1}\ K^{-1}}$$

$$\mathrm{C_v} = \mathrm{C_p} - \mathrm{R} = 7 - 2 = 5\mathrm{calmol^{-1}\ K^{-1}}$$

Amount of heat required at constant valume

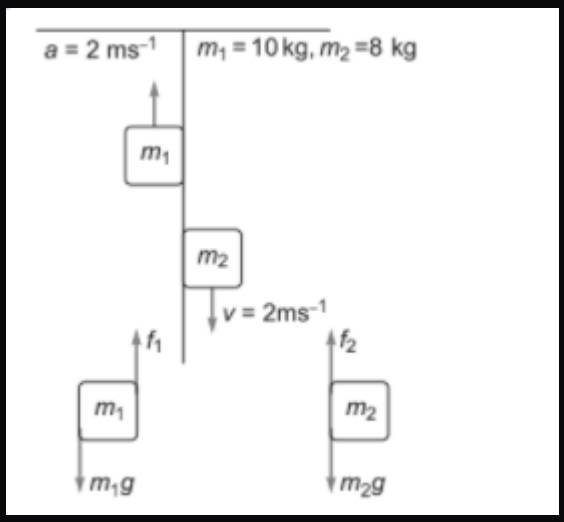
$$\mathrm{Q} = \mathrm{nC_v}\Delta\theta = 2 \times 5 \times 5 = 50\mathrm{cal}$$

Q.22 Answer:

—

Solution:

Since, m_2 moves with constant velocity



$\therefore \quad f_2 = m_2 g$

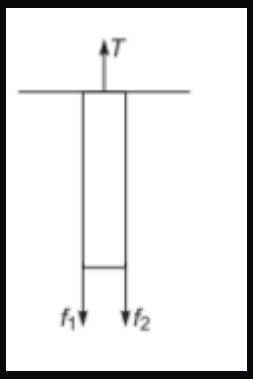
$f_2 = 8 \times 10 = 80\ \mathrm{N}$

Since, body of mass m_1 moves with acceleration $\mathrm{a} = 2\ \mathrm{ms^{-2}}$ in upward direction.

$$\therefore \quad f_1 = m_1 g + m_1 a$$
$$= 10 \times 10 + 10 \times 2$$
$$= 120\ \mathrm{N}$$

$\therefore$ Tension in string $= f_1 + f_2$

$= 120 + 80 = 200$ N



Q.23 Answer:

—

Solution:

The horizontal force $\mathbf{F}$ imparts a linear impulse $I = \int F dt =$ change in linear momentum

$\Rightarrow I = Mv - 0 = Mv = MR\omega$ where v is the velocity of the centre of mass of the ball.

Since it rolls without slipping, $v = R\omega$, where ω is the angular velocity.

The torque due to $\mathbf{F}$ imparts an angular impulse $J = Ih =$ change in angular momentum $= I\omega - 0 = I\omega$

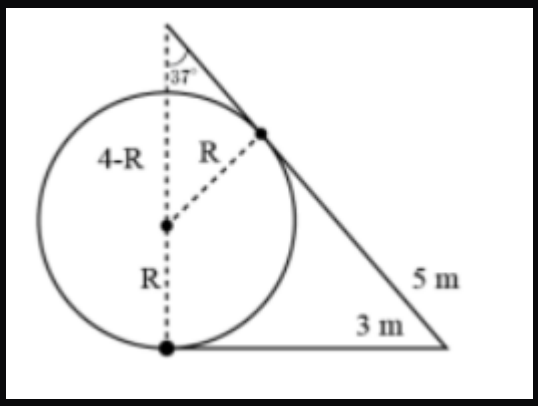
$Ih = \left(\frac{2}{5}MR^2 \right)\omega \quad \left(\begin{matrix} \because I = \frac{2}{5}MR^2 \\ \frac{2}{5}MR^2\omega \end{matrix} \right)$

Dividing (ii) by (i) $h = \frac{\frac{2}{5}MR^2\omega}{MR\omega} = \frac{2R}{5} = \frac{2 \times 2.5 \text{ cm}}{5} = 1.0$ cm

Q.24 Answer:

—

Solution:



$\sin 37^\circ = \frac{R}{4-R}$

$\Rightarrow \frac{3}{5} = \frac{R}{4-R}$

$\Rightarrow 12 - 3R = 5R$

$\Rightarrow 8R = 12$

$\Rightarrow R = \frac{12}{8} = \frac{3}{2}$ m

$V = \frac{qBR}{m} = \frac{1 \times 2 \times 1.5}{1} = 3 \text{ m s}^{-1}$

Q.25 Answer:

—

Solution:

We have $I = \frac{dq}{dt} = 3t^2 + 2t + 5$

$\Rightarrow dq = \left(3t^2 + 2t + 5\right) dt$

$\therefore q = \int_0^2 \left(3t^2 + 2t + 5\right) dt$

$= \left(\frac{3t^3}{3} + \frac{2t^2}{2} + 5t\right)^2$

$= \left[\frac{3(2)^3}{3} + 2\frac{(2)^2}{2} + 5 \times 2\right] = 8 + 4 + 10 = 22\ C$

Q.26 Answer:

It remains the same.

Solution:

The given equilibrium reaction is $\mathrm{CaCO_3(\ s) \rightleftharpoons CaO(s) + CO_2(\ g)}$.

Apply the law of chemical equilibrium,

$\mathrm{K_P = P_{CO_2}}$

For pure solids and liquids, the concentration remains unchanged during the course of the reaction and their active masses are taken as unity. Thus, adding calcium carbonate will not affect the given equilibrium.

Q.27 Answer:

$\mathrm{CuF_2}$

Solution:

In $\mathrm{CuF_2}$, $\mathrm{Cu^{2+}}$ ion exist, having d^9 configuration. Unpaired electron causes colour ($d - d$ transition). In the crystalline form, $\mathrm{CuF_2}$ is blue coloured.

and also paramagnetic due to presence of unpaired electron.

While in other compounds Cu is in positive one oxidation state and does not have unpaired electron.

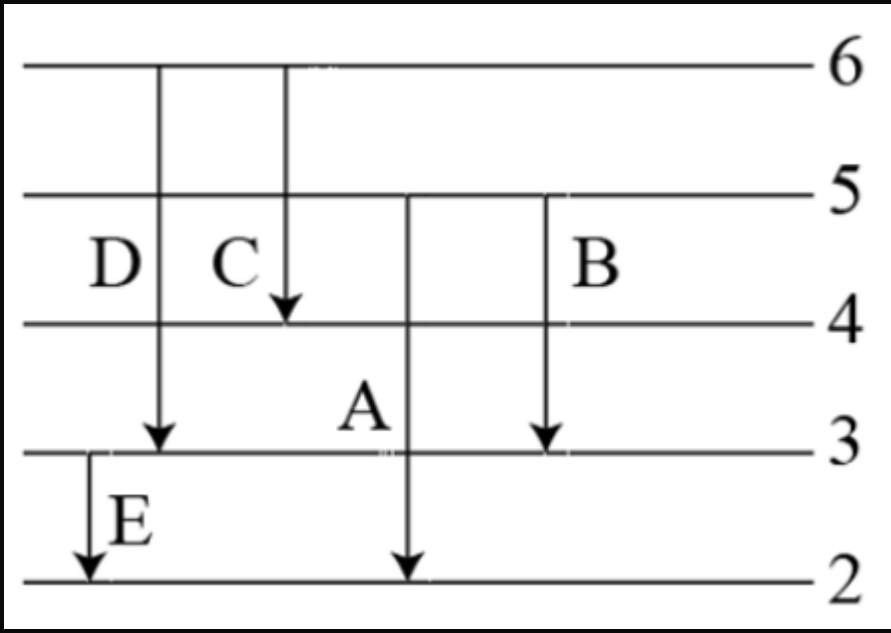
Hence all are colourless.

Q.28 Answer:

A only

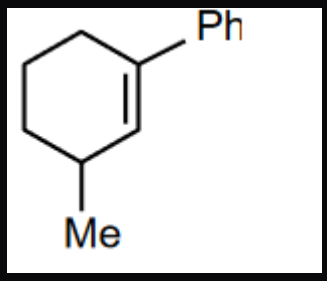
Solution:

Line E belong to visible so n_1 will be 2



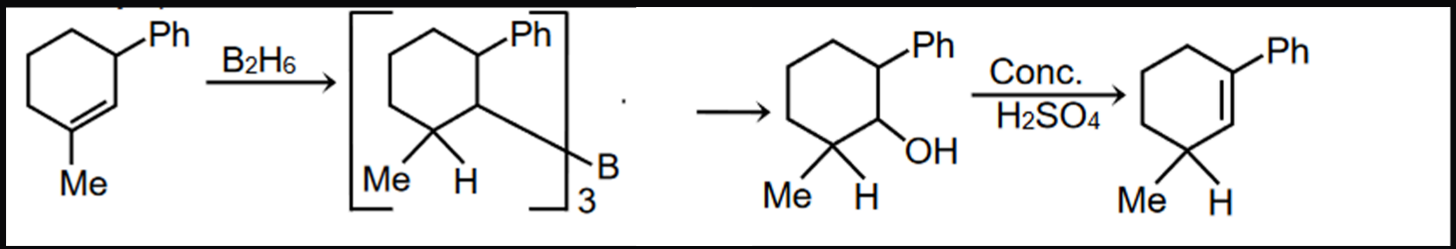
For E, $\Delta E = +13.6 \times z^2 \left(\frac{1}{4} - \frac{1}{9}\right) = 13.6 \times z^2 \times 0.138$
For D, $\Delta E = 13.6 \times z^2 \left(\frac{1}{9} - \frac{1}{36}\right) = 13.6 \times z^2 \times 0.083$
For C, $\Delta E = 13.6 \times z^2 \left(\frac{1}{16} - \frac{1}{36}\right) = 13.6 \times z^2 \times 0.034$
For A, $\Delta E = 13.6 \times z^2 \left(\frac{1}{4} - \frac{1}{25}\right) = 13.6 \times z^2 \times 0.21$
For B, $\Delta E = 13.6 \times z^2 \left(\frac{1}{9} - \frac{1}{25}\right) = 13.6 \times z^2 \times 0.07$
For ΔE for A is larger than that of E, so it means line A belong to U. V. region.

Q.29 Answer:

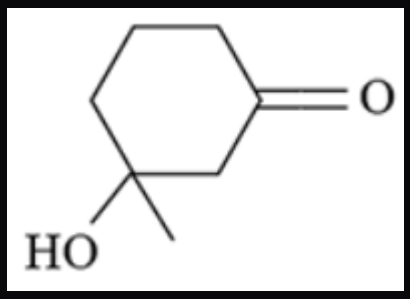


Solution:

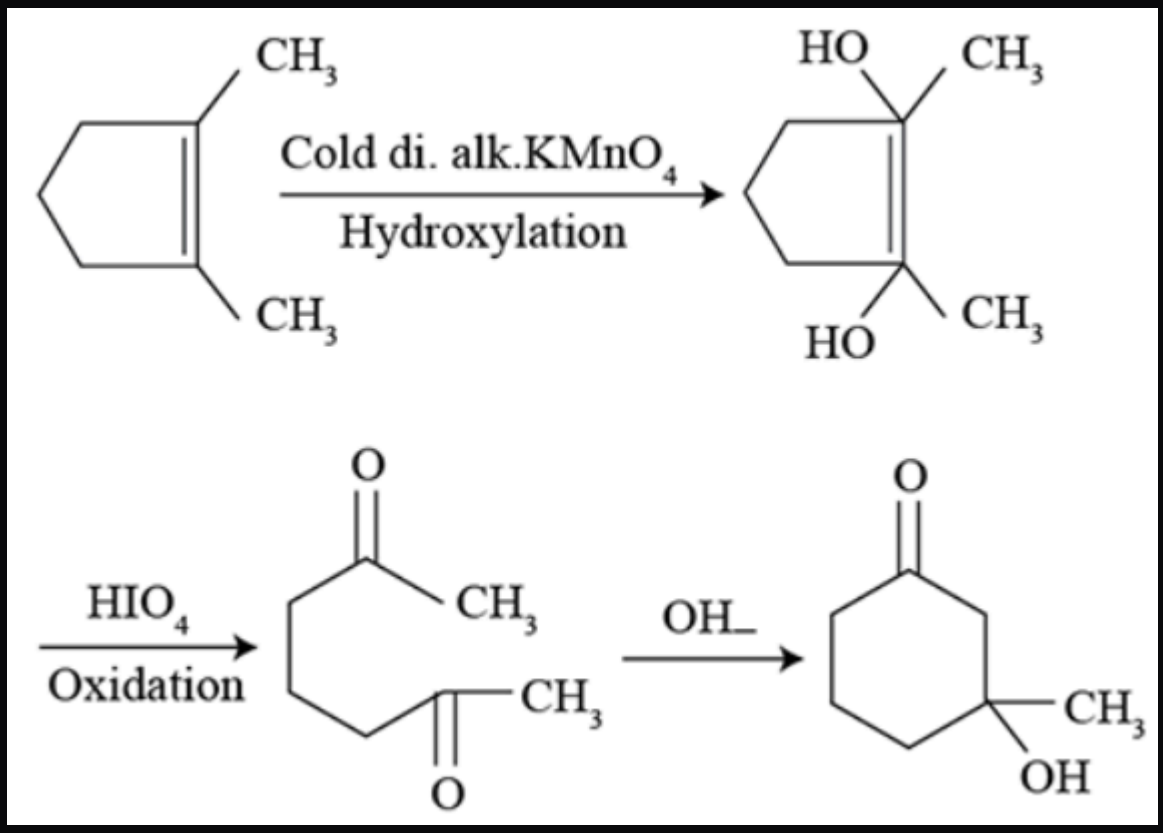
The major product of



Q.30 Answer:



Solution:

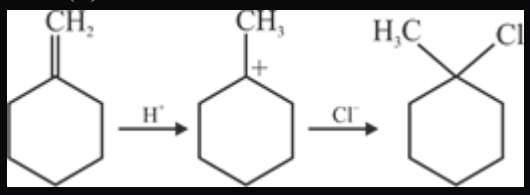


Q.31 Answer:

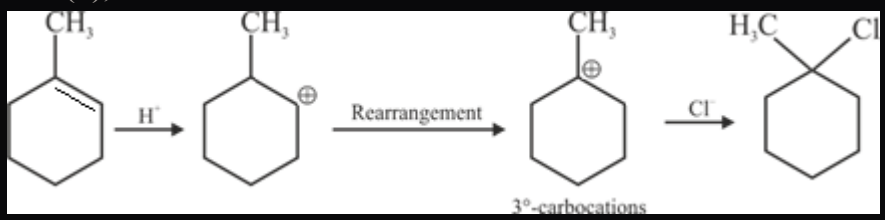
Both (A) and (B)

Solution:

For (a) :



For (b), since 3^0 carbocations are more stable than 2^0 carbocations, rearrangement will take place.



Q.32 Answer:

$[CrCl_2\,(H_2O)_4]\,Cl \cdot 2H_2O$

Solution:

$CrCl_3 \cdot 6H_2O$ has $3-Cl$ atoms but $\frac{1}{3}Cl$ atoms precipitated so one $1Cl$ atom get precipitated.
According to Werner's theory, only those ions are precipitated which are attached to the metal atoms with ionic bonds and are present outside the co-ordination sphere.

$[Cr(H_2O)_4Cl_2]Cl(aq) \rightleftharpoons [Cr(H_2O)_4Cl_2]^+(aq) + Cl^-(aq)$

$Ag^+ + Cl^- \longrightarrow AgCl\downarrow (\text{ white })$

Q.33 Answer:

0.67

Solution:

Meq. of $NaOH$+ Meq. of $Ba(OH)_2$ = Meq. of HCl

$X.V_1 + 2Y.V_2 = 100 \times 0.1 = 10$

or $4Y.\frac{V_2}{4} + 2Y.V_2 = 10$

or $3YV_2 = 10$

$2Y.V_2 = 10 \times \frac{2}{3} = 6.66$

Fraction used by $Ba(OH)_2 = \frac{6.66}{10} = 0.67$

Q.34 Answer:

Only III

Solution:

Adenine and guanine are purines whereas thymine and cytosine are pyrimidines.

Q.35 Answer:

$\frac{nA_0}{n+1}$

Solution:

$[A]_{left} = [B]_{formed} = n \times [A]_{decayed}$

$A_0e^{-\lambda t} = n \times A_0\left(1 - e^{-\lambda t}\right)$

So $e^{-\lambda t} = \frac{n}{n+1}$

Hence $[B]_{formed} = n \times A_0 \times \left(1 - e^{-\lambda t}\right)$

$= n \times A_0 \times \left(1 - \frac{n}{n+1}\right) = \frac{nA_0}{n+1}$

Q.36 Answer:

2 only

Solution:

Based on the total number electrons we can identify the atomic number of element, group number and period number in the periodic table.

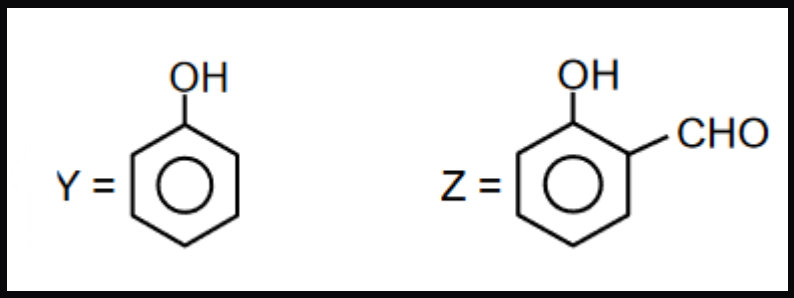
The electronic configuration $1\ s^2 2\ s^2 2p^5$ represents the element fluorine. Fluorine is the most electronegative element in the periodic table, and it belongs to 2nd period.

The electronic configuration $1\ s^2 2\ s^2 2p^3$ represents the element nitrogen. It belongs to 2nd period and 15th group in the periodic table.

The electronic configuration $1\ s^2 2\ s^2 2p^6 3\ s^2 3p^6 3\ d^8 4\ s^2$ represents the element nickel. It is a d-block element.

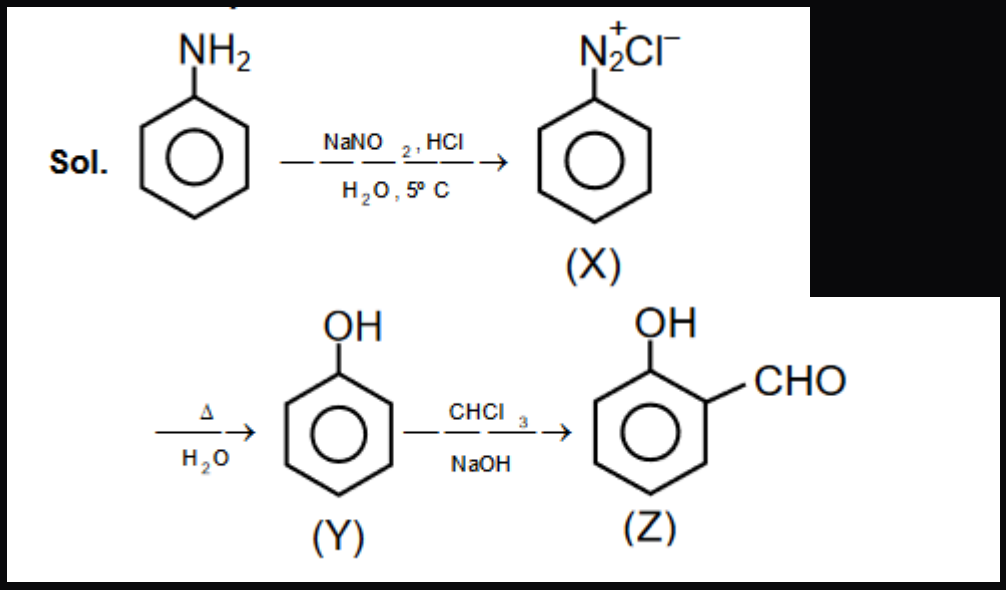
The electronic configuration $1\ s^2 2\ s^2 2p^6 3\ s^2 3p^6$ represents the element argon. It is an element from 18th group.

Q.37 Answer:



Solution:

Identify the Y and Z.



Q.38 Answer:

4.0×10^{-4} mol

Solution:

Partial pressure of N_2 in 5 atm pressure, $p_{N_2} = (0.8)(5\ \text{atm}) = 4\ \text{atm}$

By definition, Henry's law is $p_{N_2} = K_H x_{N_2}$. This gives $x_{N_2} = \frac{p_{N_2}}{K_H} = \frac{4\ \text{atm}}{1.0 \times 10^5\ \text{atm}} = 4.0 \times 10^{-5}$

Since $x_{N_2} = n_{N_2} / (n_{N_2} + n_{H_2O})$, in 10 mol of water, the amount of N_2 will be given by the expression $4.0 \times 10^{-5} = \frac{n_{N_2}}{n_N + (10\ \text{mol})}$. Solving for n_{N_2}, we get $n_{N_2} = \frac{4.0 \times 10^{-5}(10\ \text{mol})}{1 - 4.0 \times 10^{-5}} = 4.0 \times 10^{-4}$ mol

Q.39 Answer:

both (A) and (R) are true and (R) is the correct explanation of (A).

Solution:

A compound is alkaline in nature when it readily produces hydroxyl ion in aqueous solution.

$CH_3COONa + H_2O \longrightarrow CH_3COOH + NaOH$

The above process takes place in following steps:

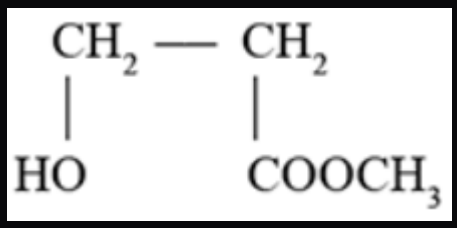
$$CH_3COONa \xrightarrow[\text{in aqueous solution}]{\text{ionisation}} \underset{\text{strong base}}{CH_3COO^-} + Na^+$$

$$CH_3COO^- + H_2O \longrightarrow CH_3COOH + OH^-$$

Acetate ion undergoes hydrolysis and the resulting solution is slightly basic, due to excess of OH^- ions.

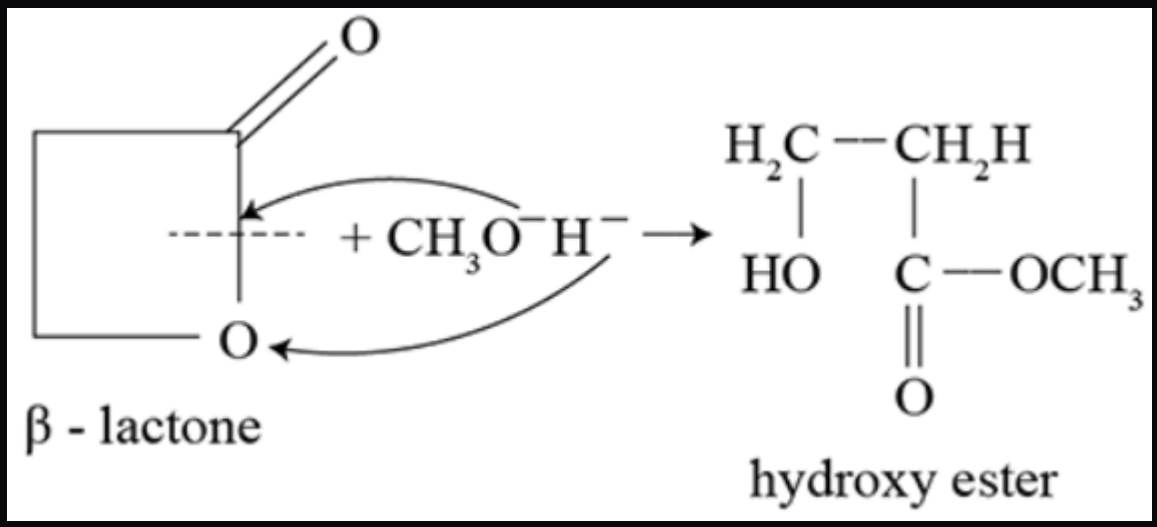
Hence, both (A) and (R) are true and (R) is the correct explanation of (A).

Q.40 Answer:



Solution:

The reaction produced as



β - lactone do not exist but can only be made by special method.

Q.41 Answer:

2,4

Solution:

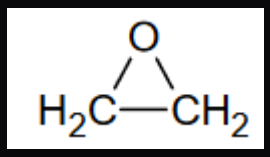
(A) Manganese show pale purple colour in flame test.

(B) $Cu^{2+} \xrightarrow{H_2SHCl} CuS\downarrow$

(C) Both Cu^{2+} **and** Mn^{2+} **form precipitate with** H_2S **in basic medium.**

(D) $E^o_{Cu^{2+}} \left| Cu = +0.34\ V, E^o_{Mn^{2+}} \right| Mn = -1.18\ V$

Q.42 Answer:



Solution:

$$\mathrm{CH_2-CH(O)} \xrightarrow[(ii)\ H^+]{(i)\ CH_3MgBr} CH_3-CH_2-CH_2-OH$$

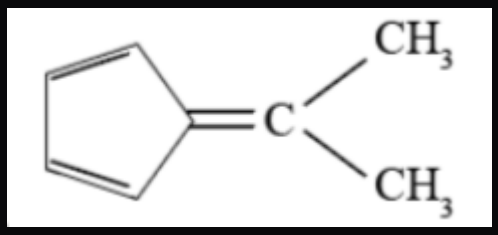
Q.43 Answer:

$(p)-(iii), (q)-(iv), (r)-(i), (s)-(ii)$

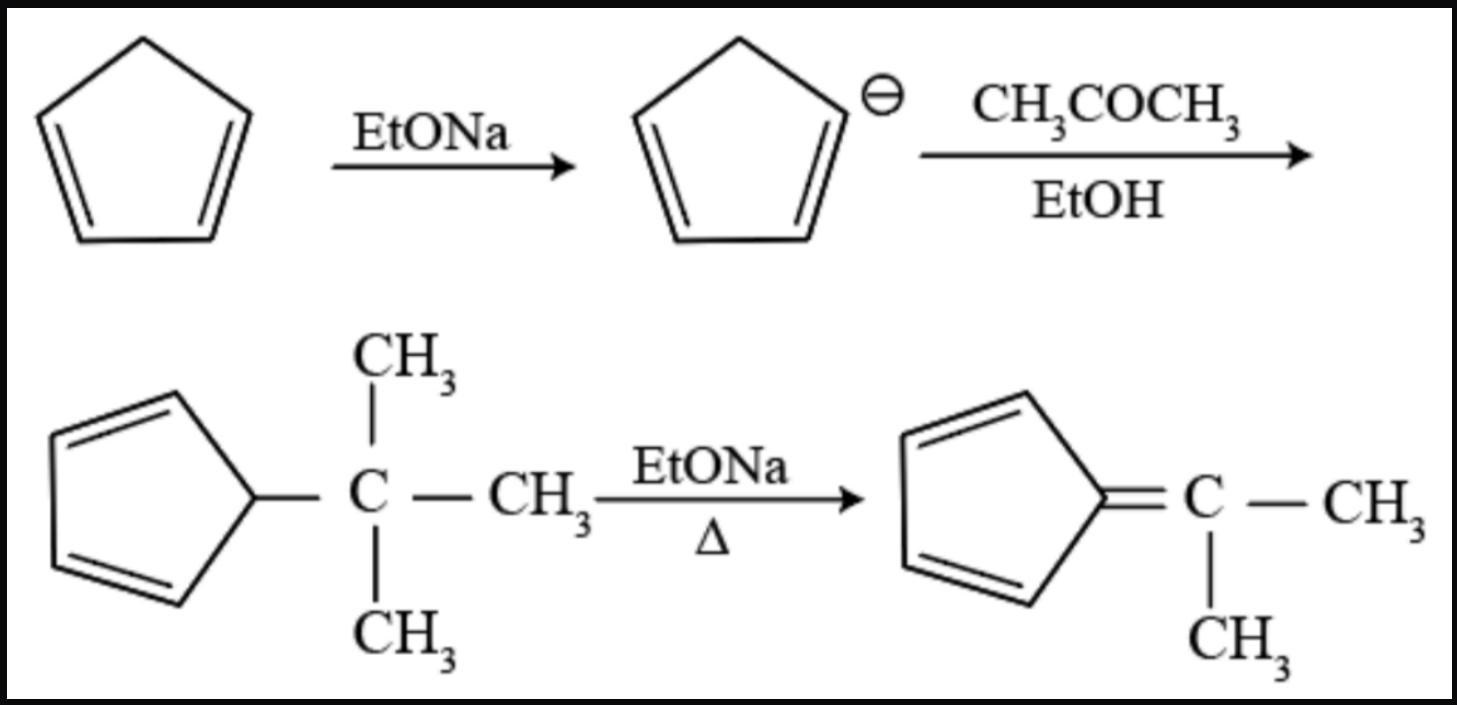
Solution:

(p) $\Delta n_g = 2-2 = 0$; hence $\Delta H = \Delta U$
(q) $\Delta n_g = 2-1 = 1$; hence $\Delta H = \Delta U + RT$
(r) $\Delta n_g = 2-4 = -2$; hence $\Delta H = \Delta U - 2RT$
(s) $\Delta n_g = 5-2 = 3$; hence $\Delta H = \Delta U + 3RT$

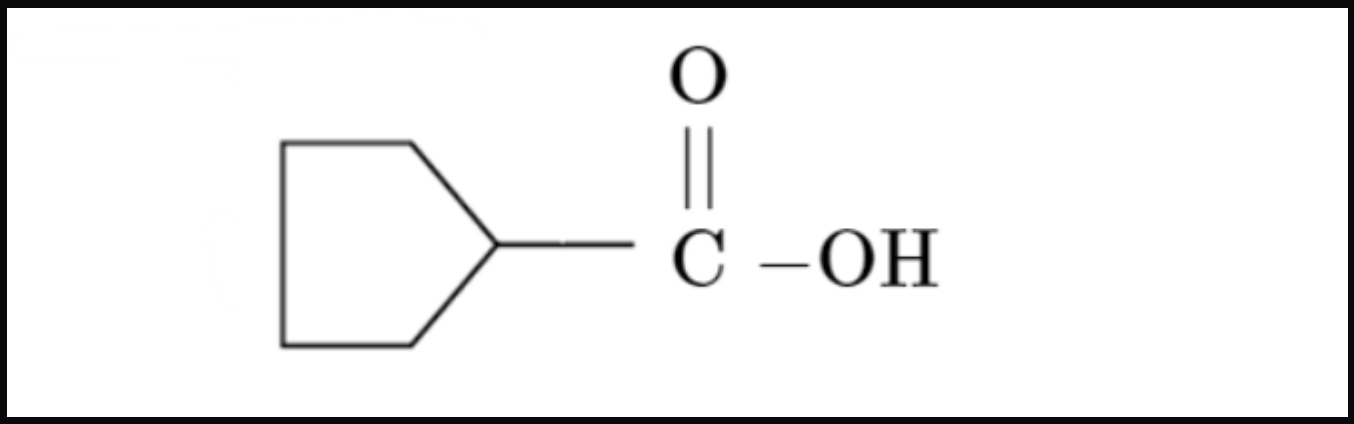
Q.44 Answer:



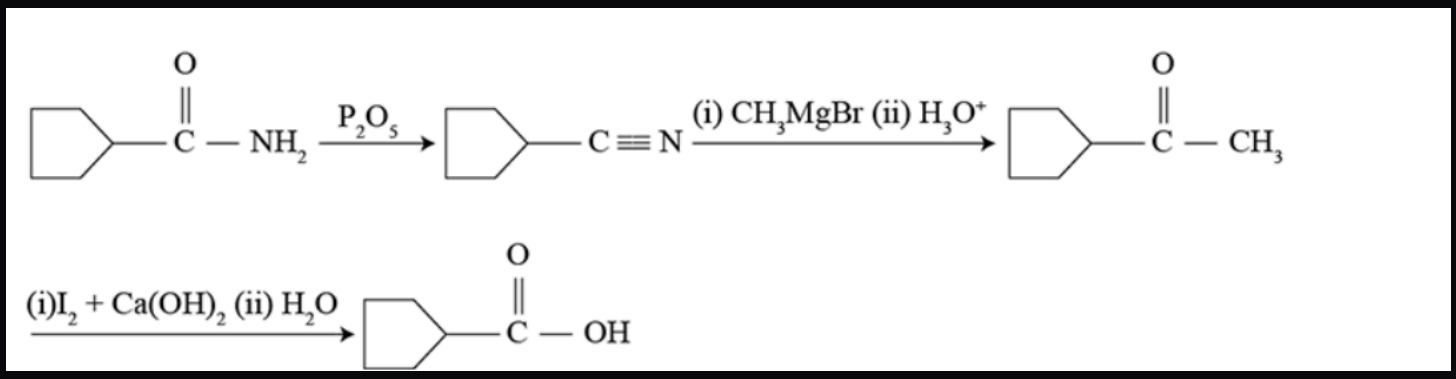
Solution:



Q.45 Answer:



Solution:



Q.46 Answer:

—

Solution:

$W = \frac{ItE}{96500}$

$E = \frac{W \times 96500}{It} = \frac{22.2 \times 96500}{2 \times 5 \times 3600} = 59.5$

Equivalent mass $= \frac{\text{Atomic mass}}{\text{Oxidation state}}$

$59.5 = \frac{177}{n}$

So $n \approx 3$.

Q.47 Answer:

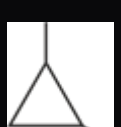
—

Solution:

Cyclic C_5H_{10}











For $3rd$ structure 2 cis-trans and 1 optical isomer are possible. Total 7 isomers.

Q.48 Answer:

—

Solution:

$8MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$

Moles of SO_4^{2-} formed from 8 moles of $MSO_4^- = 6$

Q.49 Answer:

—

Solution:

The relative lowering of vapour pressure is given by the formula:

$\frac{\Delta P}{P^\circ} = \frac{n}{n+N}$,

where, ΔP = Lowering in vapour pressure

P° = **Vapour pressure of the pure solvent**

n = **Moles of solute (Glucose)**

N = **Moles of solvent (water)**

Given,

$\Delta P = 1\%$

$N = \frac{500}{18}$

$\therefore \ \frac{1}{100} = \frac{n}{n+\frac{500}{18}} 0.01 = \frac{n}{n+27.78}$

$\Rightarrow \ n = 0.28 \ mol$

$W = n \times MW \, So, \ W = 0.28 \times 180 = 50.4 \ g$

Hence, the weight of glucose to be added is 50.4 g.

Q.50 Answer:

—

Solution:

$Energy_{photon}\left(eV\right) = \frac{1240}{wavelength(nm)}$

$Energy_{photon} = \frac{1240}{300} = 4.1 \ eV$

According to photoelectric effect, those metals will show this effect which have work function value less than the energy of photon.

So on comparing energy of photon and work function values for the given metal we can conculde that in $\mathrm{Li},\ \mathrm{Na},\ \mathrm{K}\ \&\ \mathrm{Mg}$ i.e. in 4 metals , photo emission will take place.

Q.51 Answer:

$\frac{\pi}{4}$

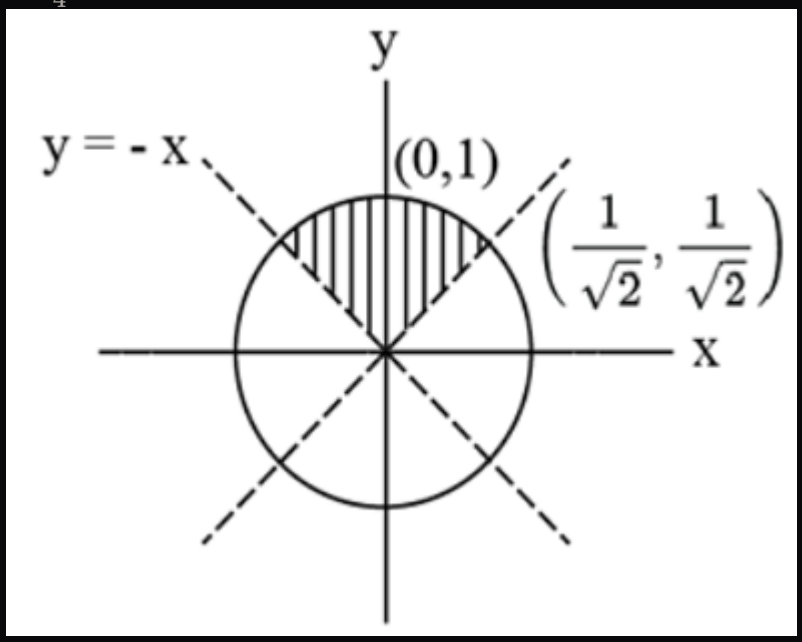
Solution:

The required area is $A = 2\int_0^{1/\sqrt{2}} \left(\sqrt{1-x^2} - x\right) dx$

$\Rightarrow A = 2\left[\frac{x\sqrt{1-x^2}}{2} + \frac{1}{2}\sin^{-1} x - \frac{x^2}{2}\right]^{1/\sqrt{2}}$

$= 2\left[\frac{1}{2\sqrt{2}}\frac{1}{\sqrt{2}} + \frac{1}{2}\frac{\pi}{4} - \frac{1}{4}\right]$

$= \frac{\pi}{4}$ sq. units

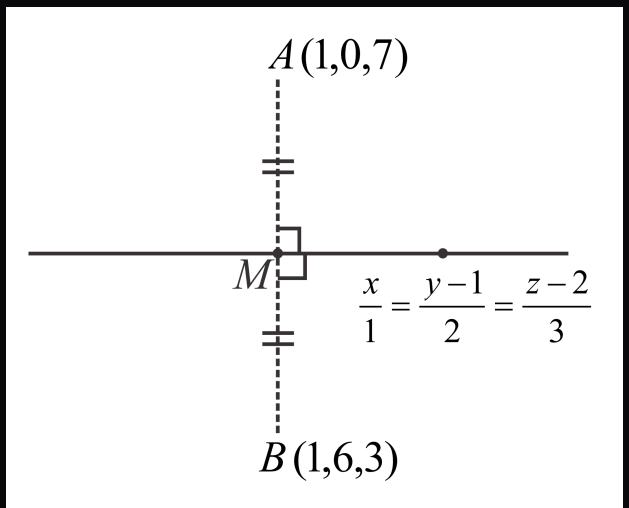


Q.52 Answer:

Statement I is true, statement II is true; statement II is a correct explanation for statement I

Solution:

Mid – point of AB is $M(1,3,5)$.



Which lies on $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

as $\frac{1}{1} = \frac{3-1}{2} = \frac{5-2}{3} \Rightarrow 1 = 1 = 1$

Hence, statement II is true

Also, directions ratios of AB is $(1–1, 6–0, 3–7) \equiv (0, 6, –4)$ $\ldots(1)$

and direction ratios of straight line is

$(1, 2, 3)$ $\ldots(2)$

The two lines are perpendicular, if

$0(1) + 6(2) + (–4)(3) = 12–12 = 0$

Hence, statement I is true and statement II is true and statement II is a correct explanation for statement I.

Q.53 Answer:

$\frac{27}{4}$

Solution:

$\left|A^{-1} adj B^{-1} adj\left(3A^{-1}\right)\right| = |A|^{-1}\left|adj B^{-1}\right|\left|adj\left(3A^{-1}\right)\right|$

$= \frac{1}{|A|}\left|B^{-1}\right|^2 \times \left|3A^{-1}\right|^2$

$= \frac{1}{|A|} \times \frac{1}{|B|^2} \times \frac{3^6}{|A|^2}$

$= \frac{3^6}{3^3 \times 2^2} = \frac{27}{4}$

Q.54 Answer:

$2y^2 = x(\ln x)^2 + 2cx$

Solution:

$2xy\frac{dy}{dx} = y^2 + x\ln x$

$2y\frac{dy}{dx} - \frac{y^2}{x} = \ln x$

Put, $y^2 = t \Rightarrow 2y\frac{dy}{dx} = \frac{dt}{dx}$

$\frac{dt}{dx} - \frac{t}{x} = \ln x$

I.F. $= e^{\int \frac{-1}{x} dx} = e^{-1nx} = \frac{1}{x}$

The solution is $\frac{t}{x} = \int \frac{\ln x}{x} dx$

$\frac{t}{x} = \frac{(\ln x)^2}{2} + c$

$\frac{y^2}{x} = \frac{(\ln x)^2}{2} + c$

$2y^2 = x(\ln x)^2 + 2cx$

Q.55 Answer:

108

Solution:

Divisible by six means divisible by 2 as well as 3.

(i) Case-1: without 0 ⇒____ $\underline{2/4}$

So, we have to put 2 or 4 in ones place.

⇒ Number of ways $= \underline{4} \times \underline{3} \times \underline{2} \times \underline{1} \times \underline{2} = 2 \times 4! = 48$

(ii) Case-2: 0 at unit place and without 3

____ $\underline{0} \Rightarrow 4 \times 3 \times 2 \times 1 \times 1 = 4! = 24$

(ii) Case-2: 2 or 4 at unit place and without 3

$\underline{\quad\quad}\ \underline{2/4} \Rightarrow 3 \times 3 \times 2 \times 1 \times 2 = 36$

$\Rightarrow$ **Total number of ways** $= 48 + 24 + 36 = 108$

Q.56 Answer:

$y = 7x - 6$

Solution:

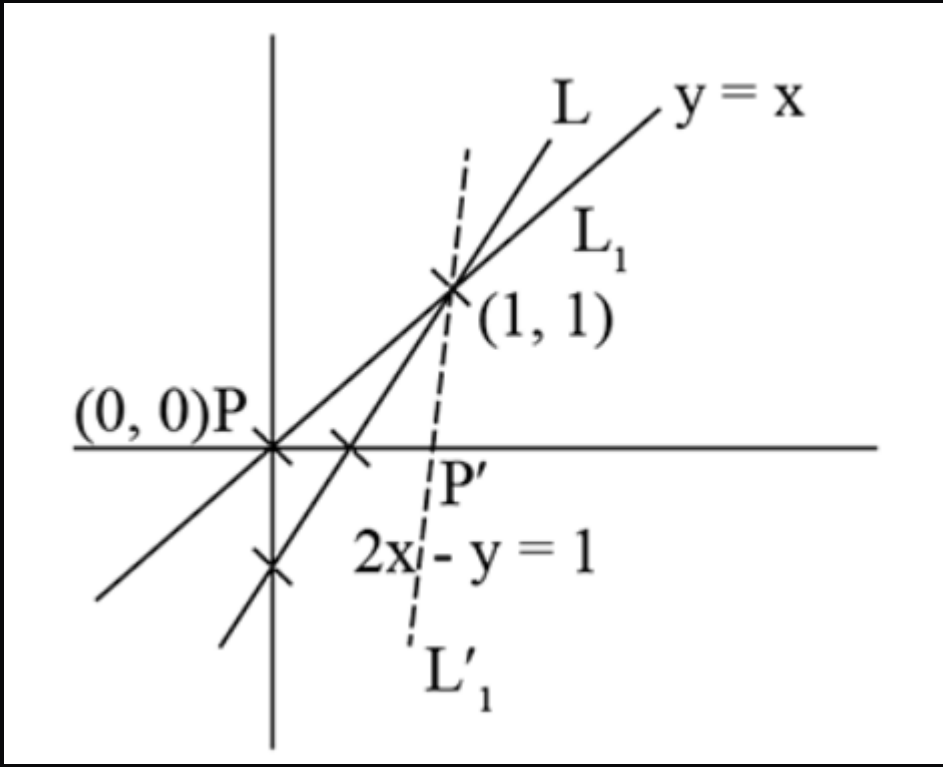


Image of $(0, 0)$ **w.r.t** L **lies on** L_1

Image of $(0, 0)$ **w.r.t** L **is**

$\frac{x-0}{2} = \frac{y-0}{-1} = -2\left(\frac{-1}{5}\right)$

$P' \equiv (x, y) = \left(\frac{4}{5}, \frac{-2}{5}\right)$

Equation of L_1' **which passes through** $(1, 1)$ **and** $\left(\frac{4}{5}, \frac{-2}{5}\right)$ **is**

$y - 1 = \frac{\frac{-2}{5}-1}{\frac{4}{5}-1}(x - 1) = \frac{-7}{-1}(x - 1)$

$y = 7x - 6$

Q.57 Answer:

3

Solution:

$f'(x) < 0$

$\therefore f\left(x^3 + f(x)\right) \leq f\left(-f(x) - x^3\right)$

$\Rightarrow x^3 + f(x) \geq -f(x) - x^3$

$\Rightarrow f(x) + x^3 \geq 0$

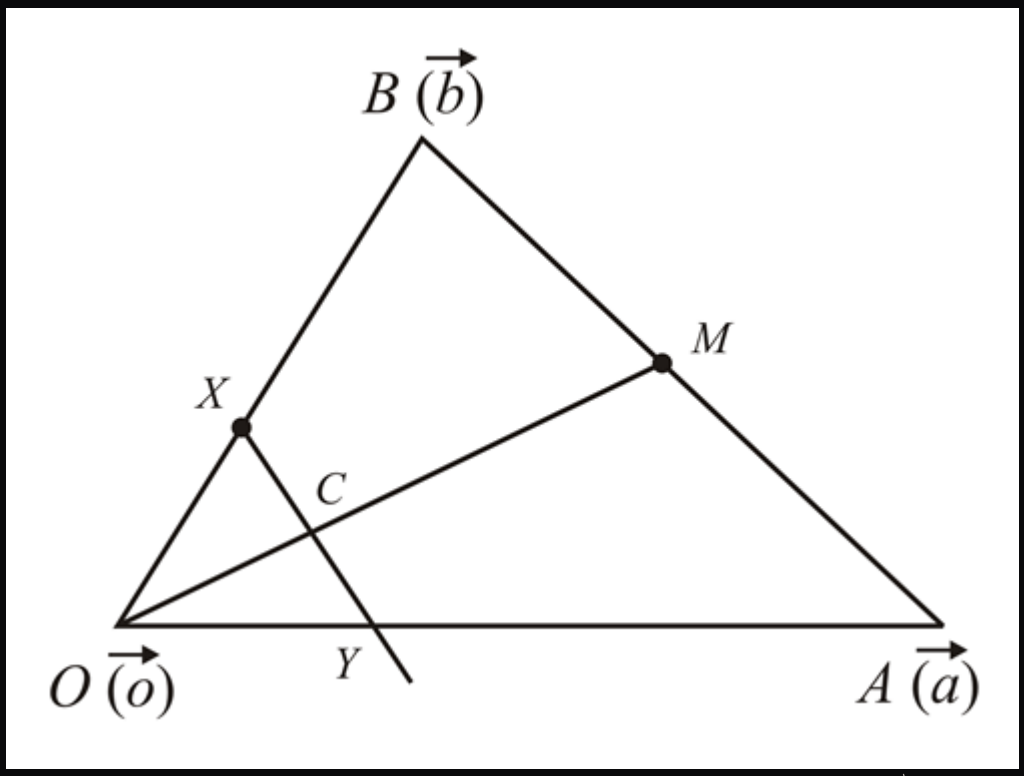
$\Rightarrow 3x^2 + 6x - 1 \leq 0$

As $x \in Z, x \in \{-2, -1, 0\}$

Q.58 Answer:

$\frac{2}{7}$

Solution:



Let O be the origin and the position vectors of A and B be $\vec{a}$ and $\vec{b}$, respectively. Then,

$\overrightarrow{OA} = \vec{a}$ **and** $\overrightarrow{OB} = \vec{b}$

M is the midpoint of AB. Hence,

$OM = \frac{\vec{a}+\vec{b}}{2}$.

Given,

$2\overrightarrow{OC} = \overrightarrow{CM}$

So, C divides OM in $1 : 2$ ratio.

$\therefore \overrightarrow{OC} = \frac{\vec{a}+\vec{b}}{6}$

Given,

$\overrightarrow{OX} = 2\overrightarrow{XB}$

So, X divides OB in $1 : 2$ ratio.

$\therefore \overrightarrow{OX} = \frac{2}{3}\overrightarrow{OB}$

$\Rightarrow \overrightarrow{OX} = \frac{2}{3}\vec{b}$

Let $\frac{OY}{YA}$ be equal to k.

So, Y divides OA in $k : 1$ ratio.

$\therefore \overrightarrow{OY} = \frac{k}{k+1}\vec{a}$

Now, the points Y, C and X are collinear.

$\therefore \overrightarrow{YC} = l\overrightarrow{CX}$**, where l is a scalar.**

$\Rightarrow \overrightarrow{OC} - \overrightarrow{OY} = l\left(\overrightarrow{OX} - \overrightarrow{OC}\right)$

$\Rightarrow \frac{\vec{a}+\vec{b}}{6} - \left(\frac{k}{k+1}\right)\vec{a} = l\frac{2\vec{b}}{3} - l\left(\frac{\vec{a}+\vec{b}}{6}\right)$.

Comparing the coefficients of $\overrightarrow{a}$ and $\overrightarrow{b}$ as both are non-collinear vectors, we get,

$\frac{1}{6}-\frac{k}{k+1}=-\frac{l}{6}$ and $\frac{1}{6}=\frac{2l}{3}-\frac{l}{6}$.

On solving, we get $l=\frac{1}{3}$ and $k=\frac{2}{7}$.

Substituting the value of k, we get

$\frac{OY}{YA}=\frac{2}{7}$.

Q.59 Answer:

$\frac{xy}{1-x-y+2xy}$

Solution:

A and B will agrec in a certain statement if both speak truth or both tell a lie. We define following events

$\mathrm{E}_1=\mathrm{A}$ and B both speak truth $\Rightarrow \mathrm{P}\left(\mathrm{E}_1\right)=\mathrm{xy}$

$\mathrm{E}_2=$ A and B both tell a lie $\Rightarrow \mathrm{P}\left(\mathrm{E}_2\right)=(1-\mathrm{x})(1-y)$

$\mathrm{E}=\mathrm{A}$ and B agree in a certain statement

Clcarly, $\mathrm{P}\left(\mathrm{E} / \mathrm{E}_1\right)=1$ and $\mathrm{P}\left(\mathrm{E} / \mathrm{E}_2\right)=1$

The required probability is $\mathrm{P}\left(\mathrm{E}_1 / \mathrm{E}\right)$.

Using Baye's theorem

$$
\begin{aligned}
& P\left(E_1 / E\right) \\
& =\frac{P\left(E_1\right) P\left(E / E_1\right)}{P\left(E_1\right) P\left(E / E_1\right)+P\left(E_2\right) P\left(E / E_2\right)} \\
& =\frac{xy.1}{xy.1+(1-x)(1-y) \cdot 1}=\frac{xy}{1-x-y+2xy}
\end{aligned}
$$

Q.60 Answer:

$\{-3,3\}$

Solution:

$\left|\mathrm{T}_3\right|<\left|\mathrm{T}_4\right|>\left|\mathrm{T}_5\right|$

$\Rightarrow\left|\frac{\mathrm{T}_4}{\mathrm{T}_3}\right|>1$ and $\left|\frac{\mathrm{T}_5}{\mathrm{T}_4}\right|<1$

$\mathrm{T}_{\mathrm{r}+1}={}^{10}\mathrm{C}_r 2^{10-r}\left(\frac{3x}{8}\right)^r$

$\frac{\mathrm{T}_4}{\mathrm{T}_3}=\frac{{}^{10}\mathrm{C}_3 2^7\left(\frac{3x}{8}\right)^3}{{}^{10}\mathrm{C}_2 2^8\left(\frac{3x}{8}\right)^2}=\frac{x}{2}$

$\frac{\mathrm{T}_5}{\mathrm{T}_4}=\frac{{}^{10}\mathrm{C}_4 2^6\left(\frac{3x}{8}\right)^4}{{}^{10}\mathrm{C}_3 2^7\left(\frac{3x}{8}\right)^3}=\frac{21}{64}x$

$\left|\frac{x}{2}\right|>1 \Rightarrow x>2$ or $x<-2$

$\left|\frac{21}{64}x\right|<1 \Rightarrow -\frac{64}{21}<x<\frac{64}{21}$

Taking intersection, we get,

$x \in \left(-\frac{64}{21},-2\right) \cup \left(2,\frac{64}{21}\right)$

Hence, integral values of $x=-3,+3$

Q.61 Answer:

$f(x)$ is differentiable in R

Solution:

We have, $f\left(\frac{x+y}{3}\right)=\frac{f(x)+f(y)}{3}, f(0)=0$ and $f'(0)=5$

$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

$=\lim_{h\to 0}\frac{f\left(\frac{3z+3h}{3}\right)-f\left(\frac{3x+0}{3}\right)}{h}$

$=\lim_{h\to 0}\frac{\frac{f(3x)+f(3h)}{3}-\frac{f(3x)+f(0)}{3}}{h}$

$=\lim_{h\to 0}\frac{f(3h)-f(0)}{3h}=f'(0)=5$

$\therefore f(x)=5x+c$

$\therefore f(0)=0\Rightarrow c=0$

Hence, $f(x)=5x$

Q.62 Answer:

$b=\frac{5\pi}{4}$

Solution:

$f(x)=\tan^{-1}x+\sin^{-1}x+\left(\sin^{-1}x+\cos^{-1}x\right)$

$f(x)=\tan^{-1}x+\sin^{-1}x+\frac{\pi}{2}; x\in[-1,1]\left\{\because \sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}\right\}$

$\because f(x)$ is an increasing function

$\therefore$ Range of $f(x)\in[f(-1),f(1)]$

$f(-1)=\tan^{-1}(-1)+\sin^{-1}(-1)+\frac{\pi}{2}=-\frac{\pi}{4}$

$f(1)=\tan^{-1}(1)+\sin^{-1}(1)+\frac{\pi}{2}=5\frac{\pi}{4}$

$\therefore$ Range of $f(x)\in\left[-\frac{\pi}{4},\frac{5\pi}{4}\right]$

Q.63 Answer:

negative

Solution:

$(2x+1)^2+(3x+2)^2+(4x+3)^2+\ldots n$ terms $=0, (x\in R)$

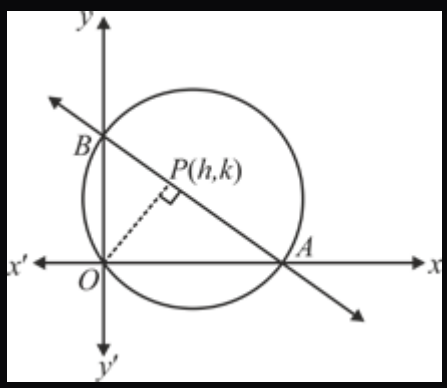
This is possible only when individually every perfect square is zero, but this can't happen here. So, the above equation doesn't have any real roots.

$\Rightarrow D<0$

Q.64 Answer:

$\left(x^2+y^2\right)^3=4R^2x^2y^2$

Solution:



Since, circle passing through origin intersect the coordinate axes at $A \& B$, hence AB must be diameter and $AB = 2R$.

Now, let foot of the perpendicular from origin upon AB be $P(h, k)$.

Slope of line $OP = \frac{k-0}{h-0} = \frac{k}{h}$

Since, line $AB \perp OP \Rightarrow$ slope of $AB = -\frac{h}{k}$

Thus, equation of line AB is $y - k = \frac{-h}{k}(x - h)$

For co-ordinates of A, put $y = 0 \Rightarrow 0 - k = \frac{-h}{k}(x - h) \Rightarrow x = \frac{h^2+k^2}{h} \Rightarrow A\left(\frac{h^2+k^2}{h}, 0\right)$

For co-ordinates of B, put $x = 0 \Rightarrow y - k = \frac{-h}{k}(0 - h) \Rightarrow y = \frac{h^2+k^2}{k} \Rightarrow B\left(0, \frac{h^2+k^2}{k}\right)$.

Now, given $AB = 2R$

Applying distance formula,

$$\Rightarrow \sqrt{\left(\frac{h^2+k^2}{h} - 0\right)^2 + \left(0 - \frac{h^2+k^2}{k}\right)^2} = 2R$$

$$\Rightarrow \left(h^2+k^2\right)^2 \left(\frac{1}{h^2} + \frac{1}{k^2}\right) = 4R^2 \Rightarrow \left(h^2+k^2\right)^3 = 4R^2h^2k^2$$

$$\Rightarrow \left(h^2+k^2\right)^2 \left(\tfrac{1}{h^2} + \tfrac{1}{k^2}\right) = 4R^2 \Rightarrow \left(h^2+k^2\right)^3 = 4R^2h^2k^2$$

Hence, locus is $\left(x^2+y^2\right)^3 = 4R^2x^2y^2$

Q.65 Answer:

$-\frac{8}{5}$

Solution:

$AA^T = 4I \Rightarrow \left|AA^T\right| = |4I| \Rightarrow |A|^2 = 64$

$\Rightarrow |A| = -8$ or 8

As, $|A| < 0 \Rightarrow |A| = -8$

$$\begin{vmatrix} a_{11}+4 & a_{12} & a_{13} \\ a_{21} & a_{22}+4 & a_{23} \\ a_{31} & a_{32} & a_{33}+4 \end{vmatrix} = |A + 4I| = \left|A + AA^T\right|$$

$= |A|\left|I + A^T\right|$

$\Rightarrow -8|A + I| = 5\lambda|A + I|$

$\Rightarrow \lambda = -\frac{8}{5}$

Q.66 Answer:

$\mathrm{I}_1 > \mathrm{I}_2$

Solution:

As $\sin x > \cos x, \quad \forall x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$\Rightarrow e^{\sin x} > e^{\cos x}$

$\therefore \frac{e^{\sin x}+1}{e^{\cos x}+1} > \frac{e^{\cos x}+1}{e^{\sin x}+1}$

$\Rightarrow \int_{\pi}^{\frac{\pi}{2}} \frac{e^{\sin x}+1}{e^{\cos x}+1} dx > \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{e^{\cos x}+1}{e^{\sin x}+1}$

$\Rightarrow \mathrm{I}_1 > \mathrm{I}_2 > 0$

Q.67 Answer:

2

Solution:

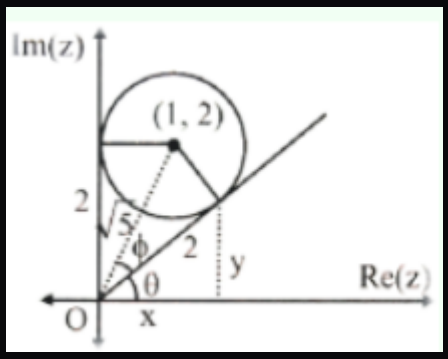
We have $2[x] = x + \{x\} = [x] + 2\{x\}$
$\Rightarrow \{x\} = \frac{[x]}{2}$
Now, $0 \leq \{x\} < 1$
$\Rightarrow 0 \leq \frac{[x]}{2} < 1$
$\Rightarrow 0 \leq [x] < 2 \Rightarrow [x] = 0, 1$
for $[x] = 0, \{x\} = 0 \Rightarrow x = 0$
for $[x] = 1, \{x\} = \frac{1}{2} \Rightarrow x = \frac{3}{2}$
Thus, $x = 0$ and $\frac{3}{2}$

Q.68 Answer:

$\frac{8}{5}$

Solution:

Here, $|z - 1 - 2i| = 1$ represents a circle with centre $(1, 2)$ and radius 1 unit.
The complex number $z = x + iy$ satisfying the given inequality and having the least positive argument is the point of contact of the tangent from the origin to the circle with the least positive slope.



From the diagram,

$$\tan\phi = \frac{1}{2}$$

$$\therefore \tan 2\phi = \frac{2\tan\phi}{1 - \tan^2\phi} = \frac{4}{3}$$

For the least positive argument, $\arg(z) = \theta$ (let)
$\tan\theta = \tan\left(\frac{\pi}{2} - 2\phi\right) = \cot 2\phi = \frac{3}{4}$
Also, from the diagram,
$x^2 + y^2 = 4$ and $\tan\theta = \frac{y}{x} = \frac{3}{4}$
i.e. $y = \frac{3}{4}x$
$\Rightarrow x^2 + \frac{9x^2}{16} = 4 \Rightarrow x = \frac{8}{5}$
Hence, for the least positive argument, the real part of z is equal to $\frac{8}{5}$

Q.69 Answer:

$\frac{9}{2}$

Solution:

$\sigma_x^2 = 3$
$\sigma_y^2 = 4$
$\bar{x} = 2$
$\bar{y} = 4$
$\frac{\Sigma x_i}{5} = 2 \Rightarrow \Sigma \mathrm{x_i} = 10; \Sigma \mathrm{y_i} = 20$
$\sigma_x^2 = \left(\frac{1}{5}\sum x_i^2\right) - (\bar{x})^2 \Rightarrow \frac{1}{5}\left(\sum x_i^2\right) - 4 = 3$
$\sum x_i^2 = 35$
$\sum y_i^2 = 100$
$\sigma_z^2 = \frac{1}{10}\left(\sum x_i^2 + \sum y_i^2\right) - \left(\frac{\bar{x}+\bar{y}}{2}\right)^2$
$= \frac{1}{10}(35 + 100) - 9 = \frac{135-90}{10}$
$= \frac{45}{10} = \frac{9}{2}$

Q.70 Answer:

44

Solution:

Given $A = \{1,\ 2,\ 3,\ 4,\ 5\}$
$f:\ A \to A$ **(into function)**
$f(i) \neq i,\ \forall\ i \in\ A$
Now number of injective functions for which
$f(i) \neq i$ **are:** $5!\left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right)$
$= 44.$

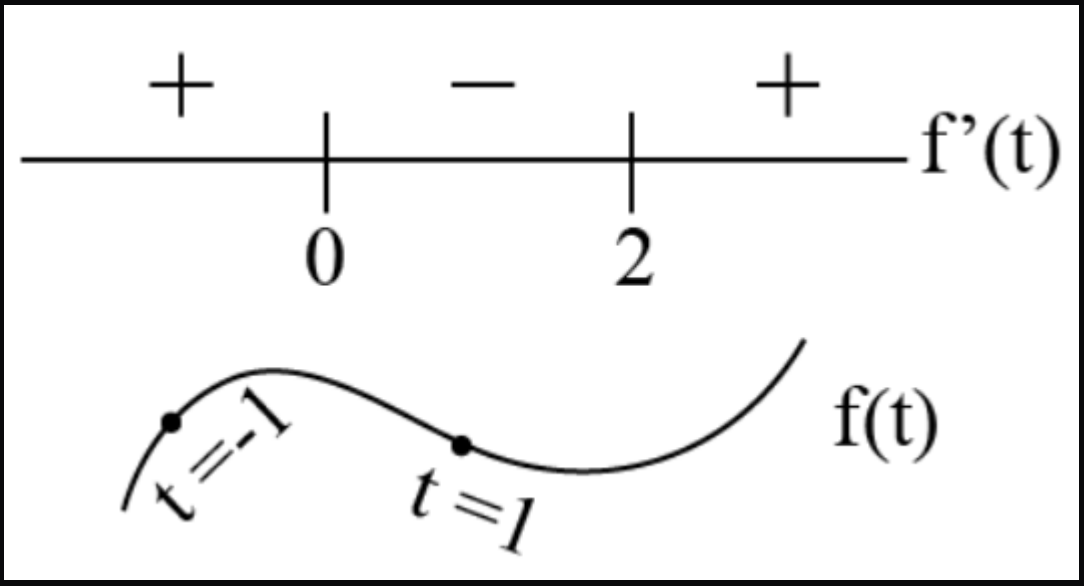
Q.71 Answer:

—

Solution:

Let, $\tan x = t \Rightarrow t \in [-1, 1] \because x \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$
$\therefore f(t) = t^3 - 3t^2 + 1$
$\Rightarrow f'(t) = 3t^2 - 6t = 3t(t-2)$
$\therefore f(-1) = -1 - 3 + 1 = -3$
$f(1) = 1 - 3 + 1 = -1$
$f(0) = 1$
$\therefore$ **Least value** $= -3$
Local maximum value $= 1$

$\therefore$ The positive difference $= 4$



Q.72 Answer:

—

Solution:

Case I $\rightarrow$ Married are invited. ${}^2C_2 \times {}^8C_3$

Case II $\rightarrow$ Married are not invited. ${}^2C_0 \times {}^8C_5$

So, No. of ways $= {}^2C_2 \times {}^8C_3 + {}^2C_0 \times {}^8C_5 = 56 + 56 = 112.$

Q.73 Answer:

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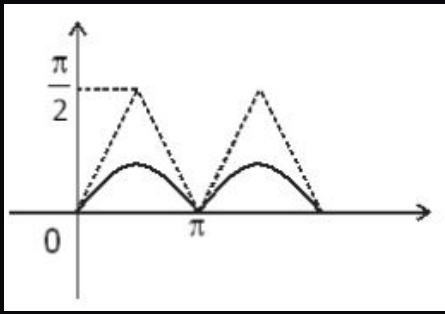
Solution:

Since $3.14 < \pi < 3.142, 1.57 < \pi/2 < 1.571$
Thus, $\left[\frac{\pi}{2} + \frac{n}{100}\right] = 1$ for $n = 0, 1, 2, \ldots, 42$
Thus, largest natural number n for which $E < 43$ is 41.

Q.74 Answer:

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Solution:



max $\left(|\sin x|, |\sin^{-1}(\sin x)|\right)$ **is** $\left|\sin^{-1}(\sin x)\right|$
Hence, $I_n = n\left(\frac{1}{2} \cdot \pi\frac{\pi}{2}\right)$
$= \frac{n\pi^2}{4}$ **(area of triangle)**
$\therefore I_2 + I_4 = \frac{6\pi^2}{4}$

Q.75 Answer:

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Solution:

Using $C_1 \to C_1 + C_2 + C_3$**, we get**

$$\Delta(x) = f(x)\begin{vmatrix} 1 & 2\cos^4 x & \sin^2 2x \\ 1 & 3 + 2\cos^4 x & \sin^2 2x \\ 1 & 2\cos^4 x & 3 + \sin^2 2x \end{vmatrix}$$

where $f(x) = 3 + 2\sin^4 x + 2\cos^4 x + \sin^2 2x$
$= 3 + 2\left(\sin^4 x + \cos^4 x + 2\sin^2 x \cos^2 x\right)$
$= 3 + 2\left(\cos^2 x + \sin^2 x\right)^2 = 5$
Applying $C_2 \to C_2 - \left(2\cos^4 x\right) C_1$ **and** $C_3 \to C_3 - \left(\sin^2 2x\right) C_1$**, we get**

$$\Delta(x) = 5\begin{vmatrix} 1 & 0 & 0 \\ 1 & 3 & 0 \\ 1 & 0 & 3 \end{vmatrix} = 45$$

Thus, $\int_{-\pi/2}^{\pi/2} x\Delta(x)dx = 45\int_{-\pi/2}^{\pi/2} xdx = 0$